

Visual Studio Code

Tips and Tricks

Foreword



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View Online neave.dev/vscode



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What is VS Code?

“ Visual Studio Code is a free code editor that is optimized for building and debugging modern web and cloud applications. ”

You can run it on Windows, Linux, Mac or Online
<https://vscode.dev>

Key Features:

- It supports a range of languages,
- Productivity shortcut circuits
 - Keyboard shortcuts
 - Tasks
 - Snippets
- Customizable with
 - Themes
 - Extensions
- Integrated terminal.

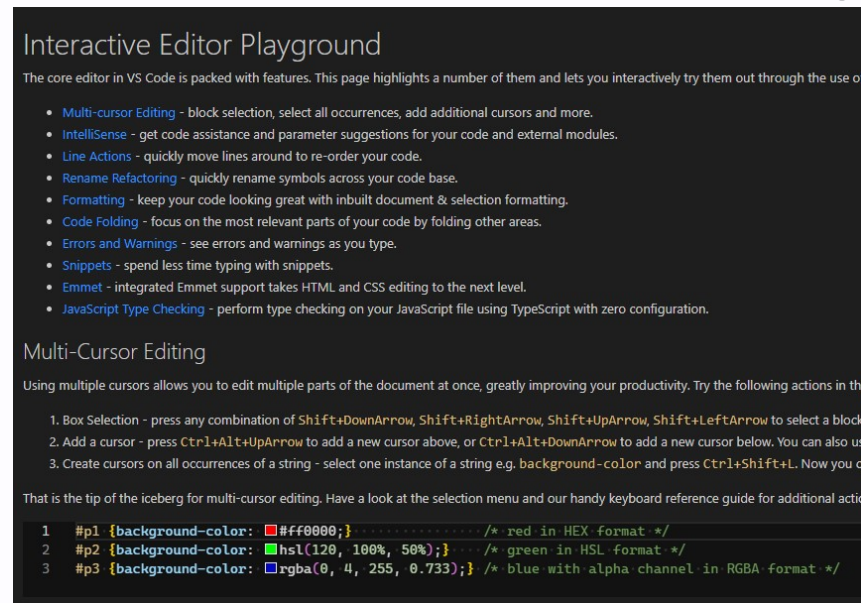
Documentation and Updates

- Official Documentation
 - Intro Videos
 - Tips and Tricks
 - Updates
- VSCode YouTube Channel

Inbuilt Features

Help Menu

- Keyboard shortcut reference
- Walkthroughs
- Interactive Editor Playground

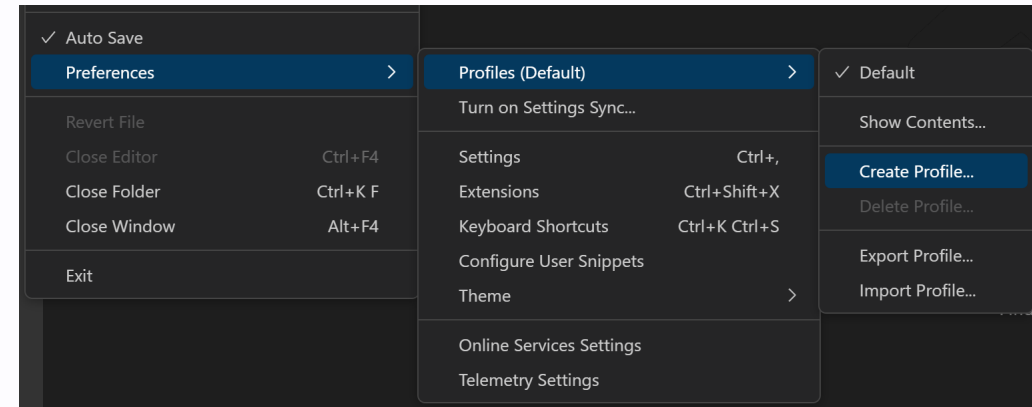


Settings

Profiles

VS Code has a Default Profile. Customize Settings, Keyboard Shortcuts, User Snippets, User Tasks or Extensions.

You can have additional
profiles.



- Have a profile for Frontend, Backend, AWS and Azure or a Project.
- Customize the profile for a stack ie (Jira, AWS and Python).
- You could install the tools and extensions as part of that project.

Customisation

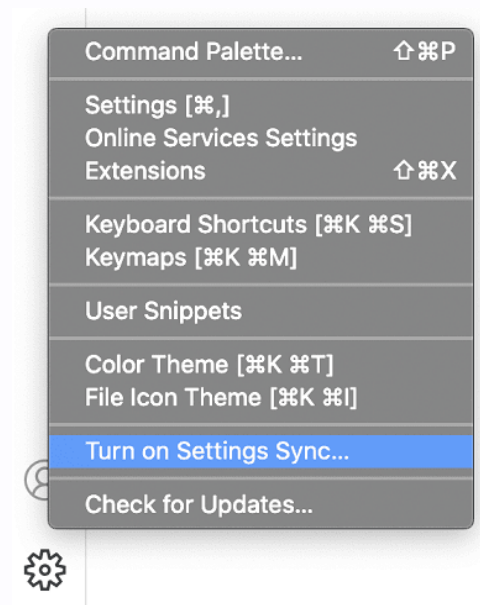
- Edit the settings via the UI or the underlying JSON.
- **Default settings** or open **Preferences: Open Default Settings (JSON)**
- Customize the User settings **Ctrl+,** (UI) - **Ctrl+P** and go to **Open User Settings (JSON)**

```
{
  "editor.fontFamily": "'Cascadia Code'",
  "editor.formatOnSave": true,
  "editor.rulers": [ 120 ],
  "editor.stickyScroll.enabled": true,
  "editor.trimAutoWhitespace": true,
  "explorer.fileNesting.enabled": true,
  "files.trimTrailingWhitespace": true,
  "git.branchPrefix": "yourusername/",
  "terminal.integrated.fontFamily": "'CaskaydiaCove Nerd Font Mono','Cascadia Code',Consolas,'Courier New',monospace",
  "workbench.sideBar.location": "right"
}
```

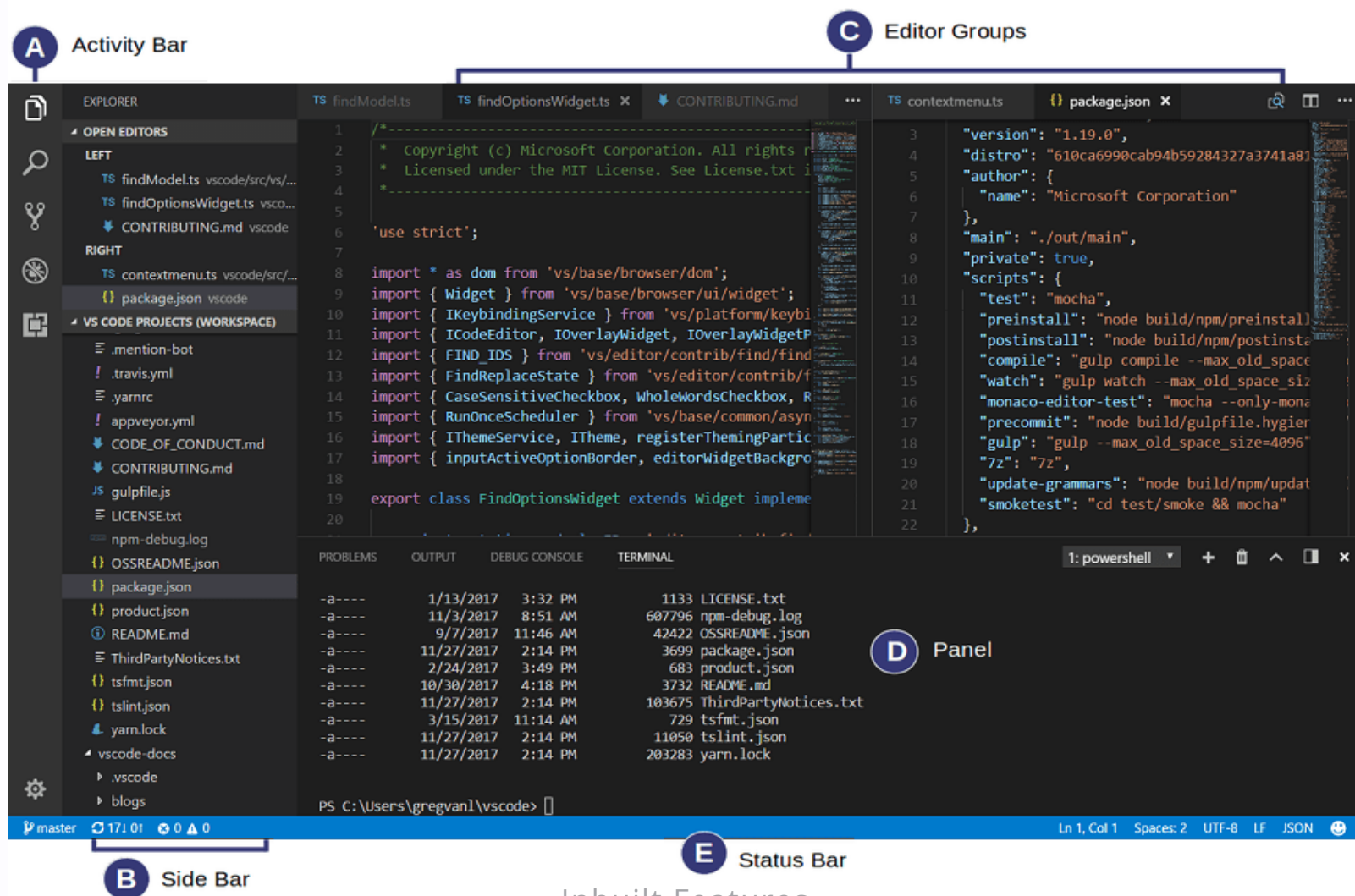
Inbuilt Features

Synchronized Settings

Settings sync across multiple instances of VS Code



Layout



Inbuilt Features

Customize the layout

- You could move the sidebar to the right then when you `Toggle Sidebar/Ctrl+B` it doesn't make the code jump around.
- Add a secondary sidebar with Outline view for better navigation.

Grouping

Ctrl+\

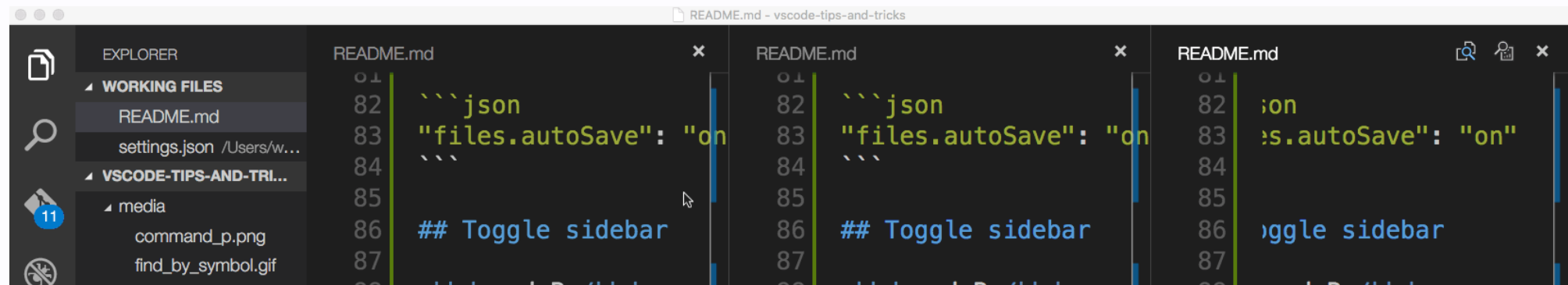
Inbuilt Features

```
elp
README.md - vscode-tips-and-tricks
README.md
31
32 ```json
33 "files.autoSave": "on"
34 ```
35
36 ## Toggle sidebar
37
38 <kbd>cmd+B</kbd>
39
40 ![toggle side bar](/media/toggle_side_bar.gif)
41
42 ## Side by side editing
43
44 <kbd>cmd+\</kbd> or <kbd>cmd</kbd> then click a file from
45
46 ## Switch between editors
47
48 <kbd>cmd+1</kbd>, <kbd>cmd+2</kbd>, <kbd>cmd+3</kbd>
49
50 ## History
51
52 Navigate entire history with <kbd>ctrl+tab</kbd>
53 Navigate back with <kbd>ctrl+-</kbd>.
54 Navigate forward with <kbd>ctrl+shift+up</kbd>
55
56 ![navigate history](/media/navigate_history.gif)
57
58 ## Navigate to a file
```

13

Spaces: 4 Ln

Switch to next group with `Ctrl+[1,2,3...]`



Alternatively: `Ctrl+P` and `View: Move Editor to Next Group`

Navigation

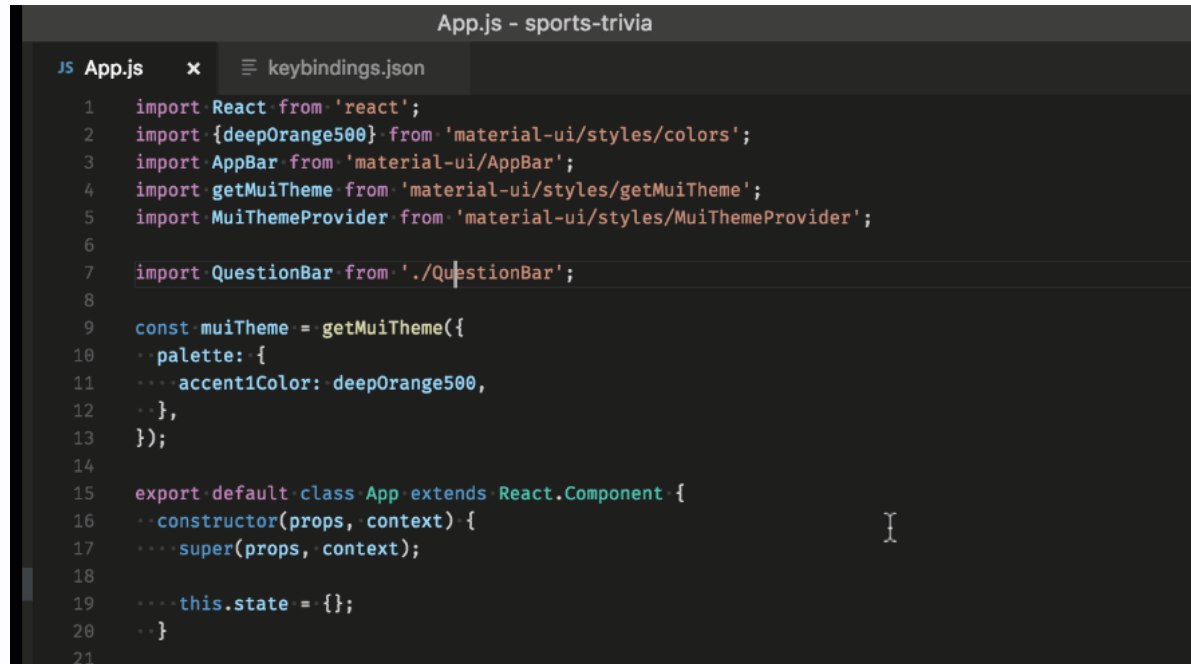
- Between projects - `Ctrl+R` Displays a Quick Pick dropdown with the list from `File > Open Recent` with recently opened folders and workspaces followed by files.
- In & around files - `Ctrl+Shift+.` to select the breadcrumb and jump.

Multi-Root Workspaces

Normally your workspace has a single root/project folder but you can have a workspace with multiple project folders in Visual Studio Code with **multi-root workspaces**. Uses a `.code-workspace` file

```
{
  "folders": [
    {
      // Source code
      "name": "Product",
      "path": "vscode"
    },
    {
      // Docs and release notes
      "name": "Documentation",
      "path": "vscode-docs"
    },
    {
      // Yeoman extension generator
      "name": "Extension generator",
      "path": "vscode-generator-code"
    }
  ]
}
```

Command Palette



The screenshot shows a code editor with a file named 'App.js - sports-trivia'. The Command Palette is open, displaying a list of commands. The first command, 'Import React from \'react\'', is highlighted. The code in the background is as follows:

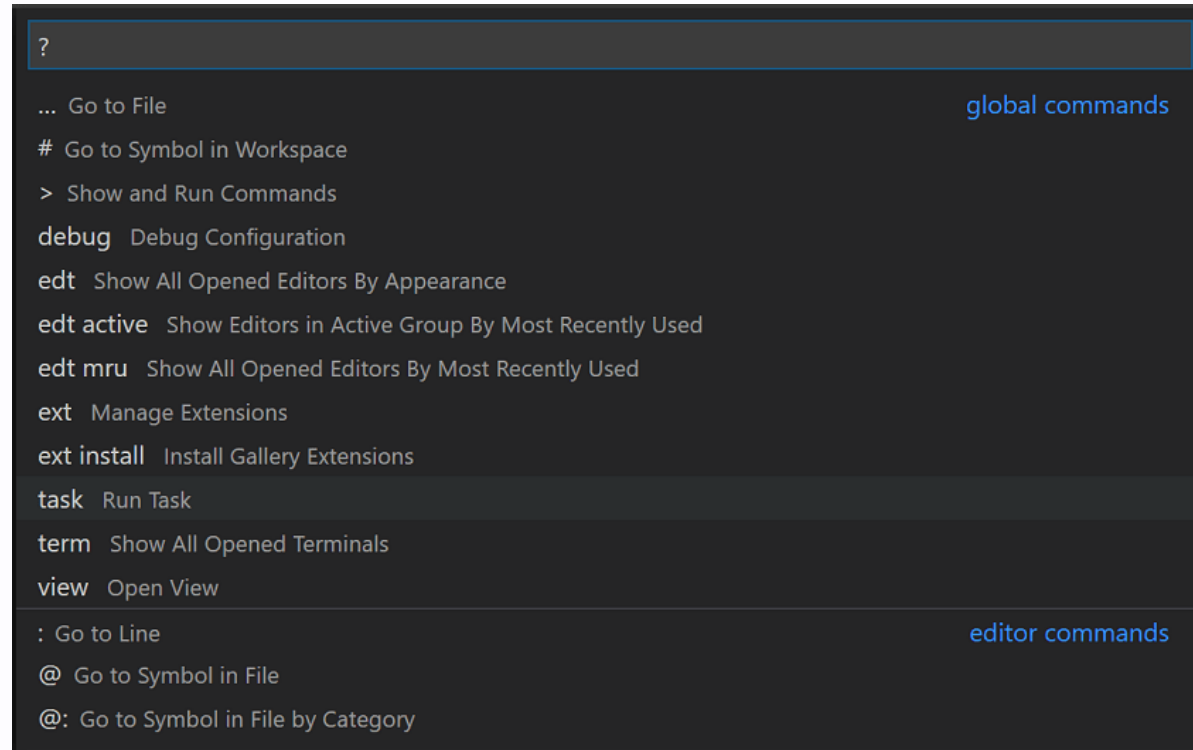
```
1 import React from 'react';
2 import {deepOrange500} from 'material-ui/styles/colors';
3 import AppBar from 'material-ui/AppBar';
4 import getMuiTheme from 'material-ui/styles/getMuiTheme';
5 import MuiThemeProvider from 'material-ui/styles/MuiThemeProvider';
6
7 import QuestionBar from './QuestionBar';
8
9 const muiTheme = getMuiTheme({
10   palette: {
11     accent1Color: deepOrange500,
12   },
13 });
14
15 export default class App extends React.Component {
16   constructor(props, context) {
17     super(props, context);
18
19     this.state = {};
20   }
21 }
```

Ctrl+Shift+P

Quick Open File



Ctrl+P



Ctrl+P and type ?term to open a new terminal

Others

`Ctrl+T` - Go To Symbol

`Ctrl+G` - Go to Line

`Alt+ ← / →` - Jump forward and back

Editing

Text Manipulation

- Sort Lines Ascending/Descending
- Join Lines
- Transform Text

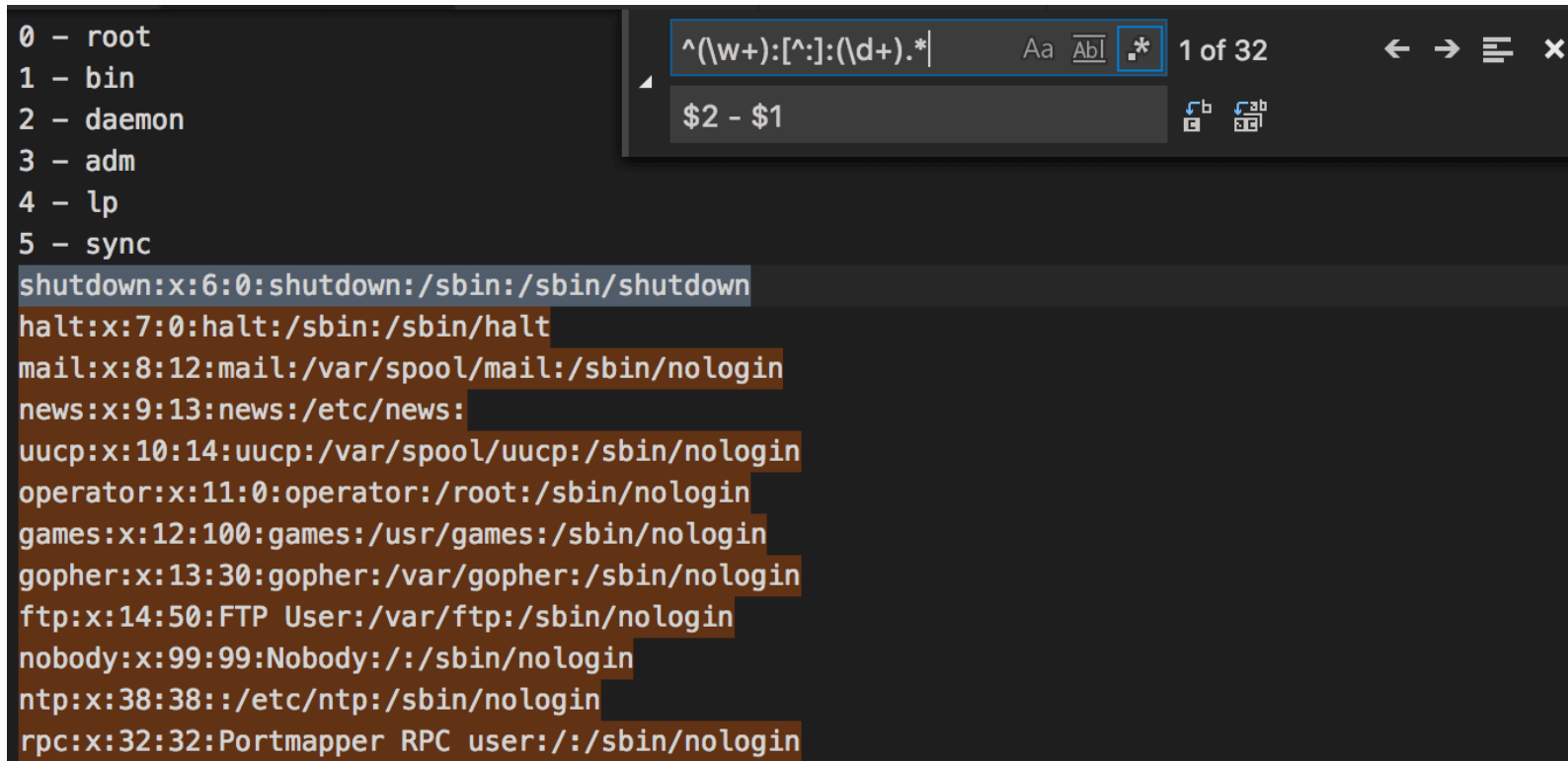
Tranform	Example
Title	Hello World
Upper	HELLO WORLD
Lower	hello world
Camel	helloWorld
Snake	helloWorld → hello_world
Kebab	helloWorld → hello-world

Code Formatting

```
render() {  
  return (  
    <div className={s.app}>  
      <AppBar  
        title="Notes"  
        iconClassNameRight="muidocs-icon-navigation-expand-more"  
      />  
      <div style={{  
        marginTop: 20,  
        marginLeft: 20  
      }}>  
        <NewNote />  
        <Notes items={this.state.notes}/>  
      </div>  
    </div>  
  );  
}
```

Regex Search and Replace

```
0 - root
1 - bin
2 - daemon
3 - adm
4 - lp
5 - sync
shutdown:x:6:0:shutdown:/sbin:/sbin/shutdown
halt:x:7:0:halt:/sbin:/sbin/halt
mail:x:8:12:mail:/var/spool/mail:/sbin/nologin
news:x:9:13:news:/etc/news:
uucp:x:10:14:uucp:/var/spool/uucp:/sbin/nologin
operator:x:11:0:operator:/root:/sbin/nologin
games:x:12:100:games:/usr/games:/sbin/nologin
gopher:x:13:30:gopher:/var/gopher:/sbin/nologin
ftp:x:14:50:FTP User:/var/ftp:/sbin/nologin
nobody:x:99:99:Nobody:/:/sbin/nologin
ntp:x:38:38:/:etc/ntp:/sbin/nologin
rpc:x:32:32:Portmapper RPC user:/:/sbin/nologin
```

A screenshot of a terminal window with a dark background. On the left, a list of system users is displayed, each on a new line. The first five lines are numbered 0 through 5, followed by a blank line, and then the names of the users: shutdown, halt, mail, news, uucp, operator, games, gopher, ftp, nobody, ntp, and rpc. Each user name is followed by a colon and then a series of fields separated by colons. The first field after the colon is the user's UID, the second is the GID, and the third is the user's full name. The fourth field is the user's home directory, and the fifth is the user's shell. The user 'shutdown' is highlighted with a blue background. A search bar is overlaid on the right side of the terminal window. It contains the regex pattern '^(\w+):[^:](\d+).*\$' and the replacement string '\$2 - \$1'. The search bar also shows '1 of 32' matches and some navigation icons.

Multi Cursor

Edit Text Vertically with [multi-cursor](#)

Keep selecting text with `Ctrl+D` and update over multiple locations in a file.

```
31 .global-message-list.transition {  
32 →   -webkit-transition: top 200ms linear;  
33 →   -ms-transition:      top 200ms linear;  
34 →   -moz-transition:     top 200ms linear;  
35 →   -khtml-transition:   top 200ms linear;  
36 →   -o-transition:       top 200ms linear;  
37 →   transition:          top 200ms linear;  
38 }
```

Expanding and Shrinking Selection

Shift+Alt+Left and Shift+Alt+Right

Find the start and end of a long `<div>` in HTML

```
7 namespace Hello1
8 {
9     0 references
10    class Program
11    {
12        0 references
13        static void Main(string[] args)
14        {
15            System.Console.WriteLine("Hello World!");
16        }
17    }
18 }
```

Folding

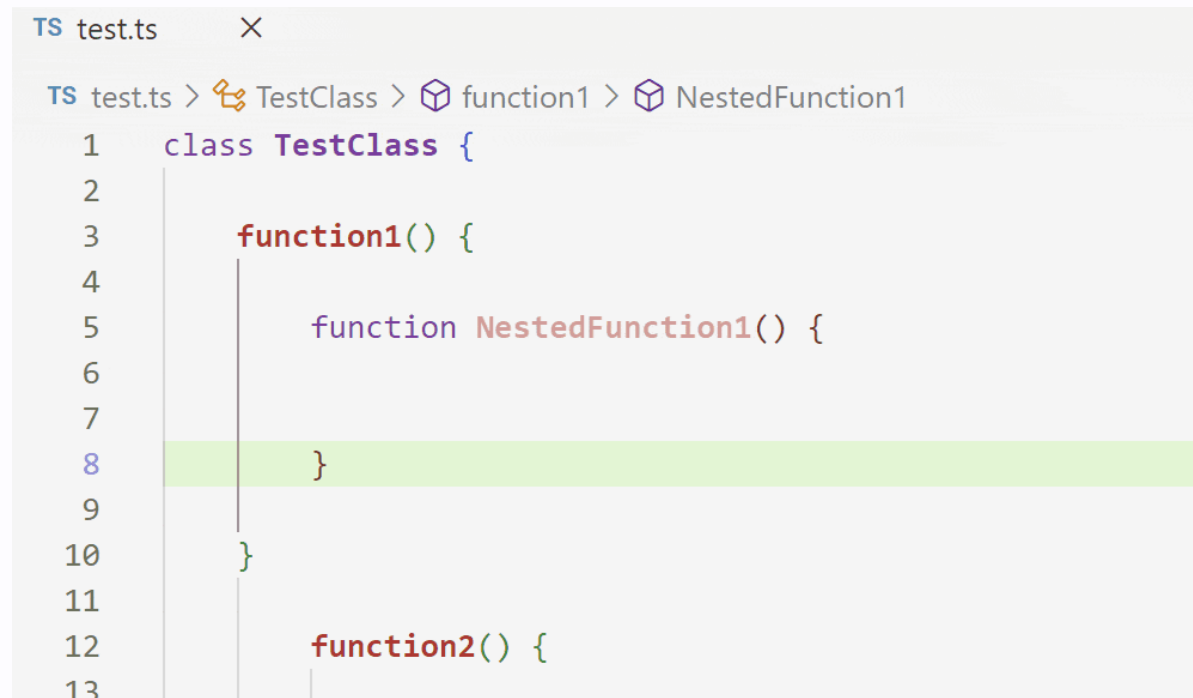
Hide Text for easier
reading

Fold Level [1 | 2 | 3 | 4],
Fold All, Unfold All

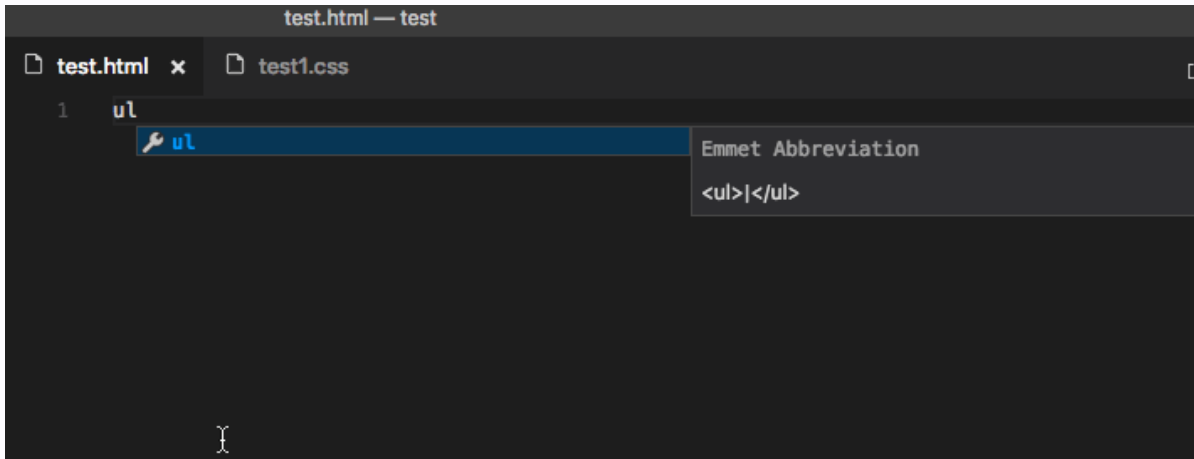
```
class App extends Component {  
  
  constructor() {  
    super();  
    this.state = {  
      notes: [  
        {  
          text: "go to the grocery store"  
        },  
        {  
          text: 'read medium article about engineering'  
        },  
        {  
          text: 'create build session'  
        },  
        {  
          text: 'fix bug #232'  
        }  
      ]  
    };  
  }  
}
```

Sticky Scroll

See the code context with Sticky Scroll - useful for long methods



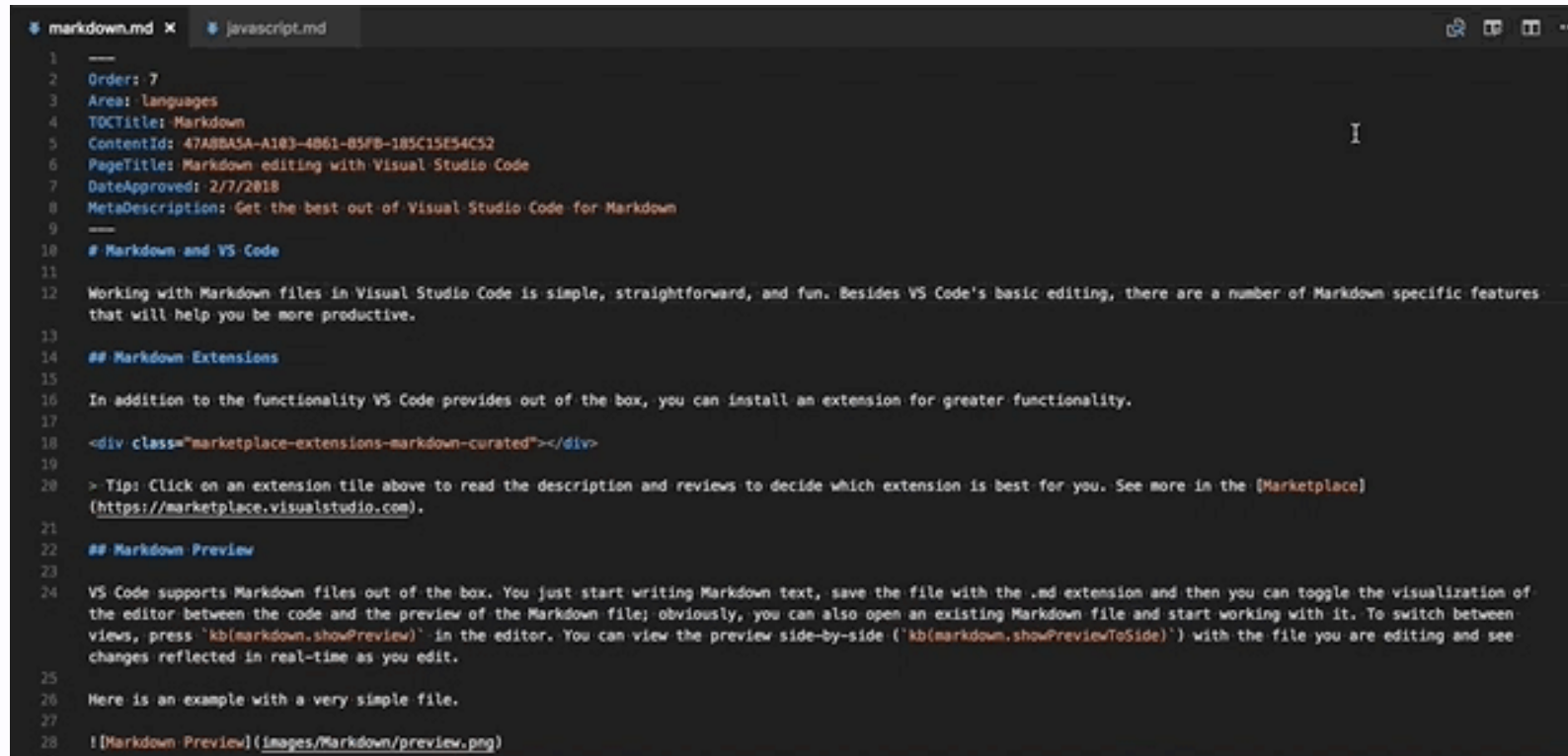
```
TS test.ts  X
TS test.ts > TestClass > function1 > NestedFunction1
1  class TestClass {
2
3      function1() {
4
5          function NestedFunction1() {
6
7
8          }
9
10     }
11
12     function2() {
13
```



Emmet

Generate HTML and
CSS from shorthand.
Works with multi-
cursors

Markdown preview



```
1 ---
2 Order: 7
3 Area: languages
4 TOCTitle: Markdown
5 ContentId: 47A8B5A-A103-4861-85FB-185C15E54C52
6 PageTitle: Markdown editing with Visual Studio Code
7 DateApproved: 2/7/2018
8 MetaDescription: Get the best out of Visual Studio Code for Markdown
9 ---
10 # Markdown and VS Code
11
12 Working with Markdown files in Visual Studio Code is simple, straightforward, and fun. Besides VS Code's basic editing, there are a number of Markdown specific features
13 that will help you be more productive.
14
15 ## Markdown Extensions
16
17 In addition to the functionality VS Code provides out of the box, you can install an extension for greater functionality.
18
19 <div class="marketplace-extensions-markdown-curated"></div>
20
21 > Tip: Click on an extension tile above to read the description and reviews to decide which extension is best for you. See more in the [Marketplace]
22 (https://marketplace.visualstudio.com).
23
24 ## Markdown Preview
25
26 VS Code supports Markdown files out of the box. You just start writing Markdown text, save the file with the .md extension and then you can toggle the visualization of
27 the editor between the code and the preview of the Markdown file; obviously, you can also open an existing Markdown file and start working with it. To switch between
28 views, press 'kb(markdown.showPreview)' in the editor. You can view the preview side-by-side ('kb(markdown.showPreviewToSide)') with the file you are editing and see
29 changes reflected in real-time as you edit.
30
31 Here is an example with a very simple file.
32
33 ! [Markdown Preview] (images/Markdown/preview.png)
34
35 ---
```

JSON Schema

Maps the JSON to a schema and provides tooltips and validation. Can be set in the

`.vscode/settings.json`

```
Defines the actions list group, providing a natural grouping of the plugin's actions.  
Note: Category should be distinctive and synonymous with your plugin, and it should  
be available under the "Custom" group.  
"Category": "Monitoring",  
"CategoryIcon": "images/plugin/category-icon",  
"CodePath": "bin/plugin.js"
```

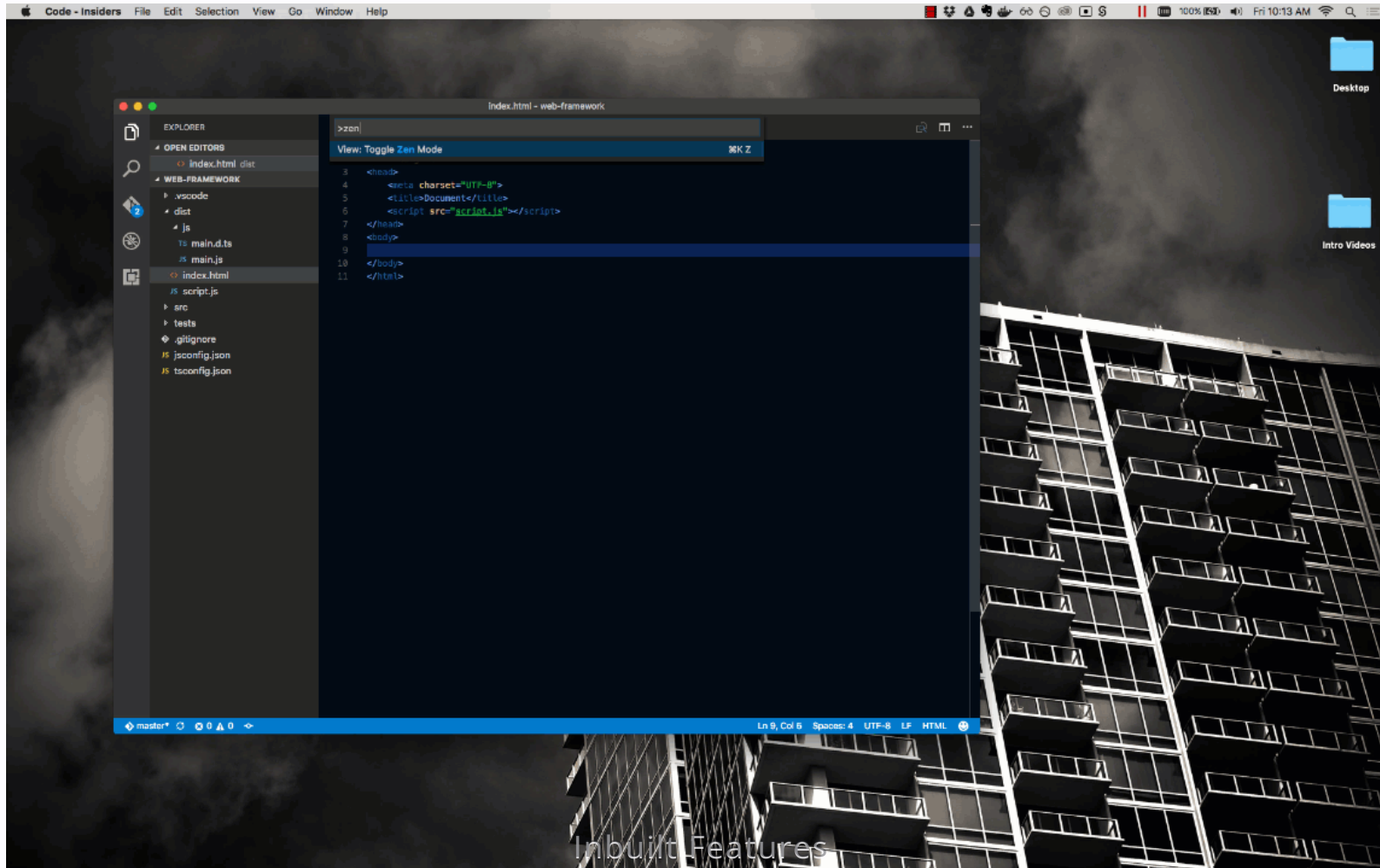
```
/* JSON schemas in settings.json */  
{  
  "json.schemas": [  
    {  
      "fileMatch": [ "**/manifest.json" ],  
      "url": "https://example.com/schemas/manifest.json"  
    }  
  ]  
}
```

Inbuilt Features

Productivity / Focus

Zen Mode

Focus on the code with Zen Mode



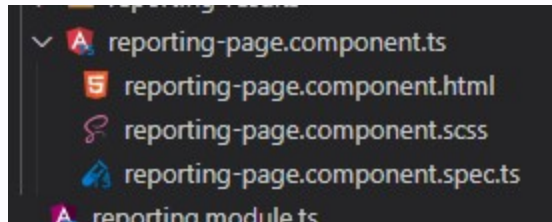
Errors

Jump to the **next error** with **F8** .

Really useful for going through compiler errors.

File Nesting

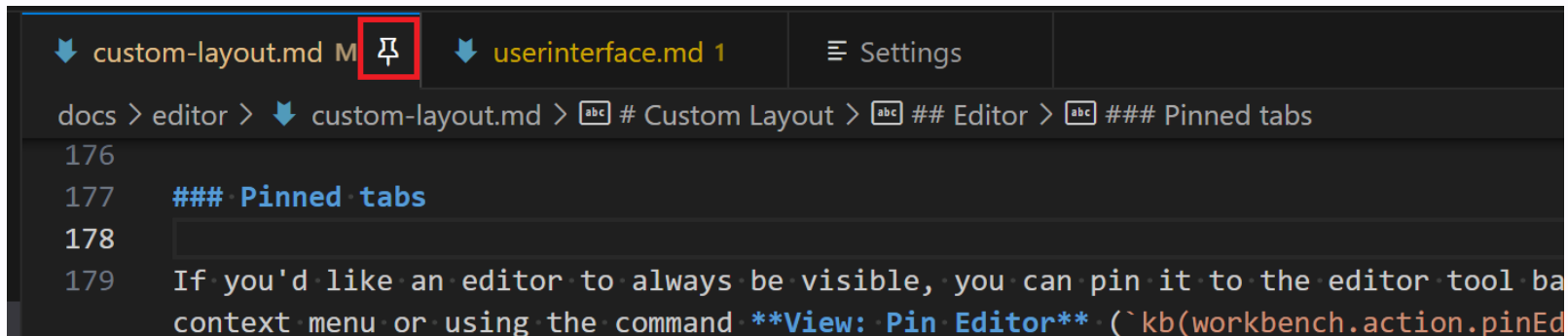
Nest related files under a parent file in the explorer.



```
"explorer.fileNesting.patterns": {  
  "*.component.ts": "$(capture).component.html, $(capture).component.spec.ts, $(capture).component.scss",  
  "README*": "AUTHORS, CHANGELOG*, CODE_OF_CONDUCT*, CONTRIBUTING*, LICENSE*"  
}
```

Pinned Tabs

Pinned Tabs allow you to keep a file open



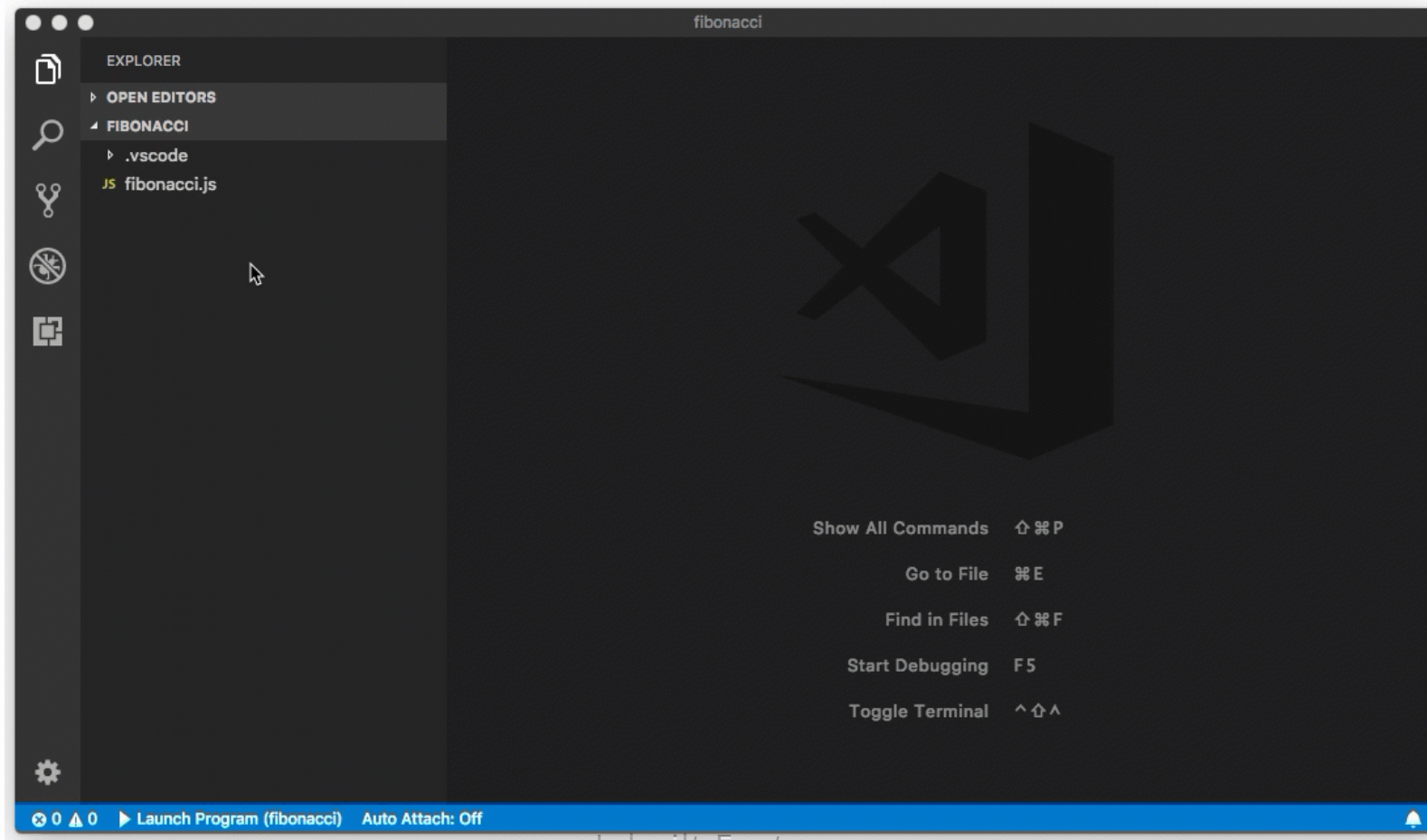
Vertical Rulers

Make sure your code isn't getting too long and hard to read.

```
JS gulpfile.js > gulp.task('build-dist') callback
20  /
21  gulp.task('build-dist', done => {
22    if (!$.which('git')) {
23      $.echo('This command requires git')
24      $.exit(1)
25      done()
26    }
27
28    // Go to _build
29    if (!$.test('-e', '_build')) {
30      $.mkdir('_build')
31    }
32    $.cd('_build')
```

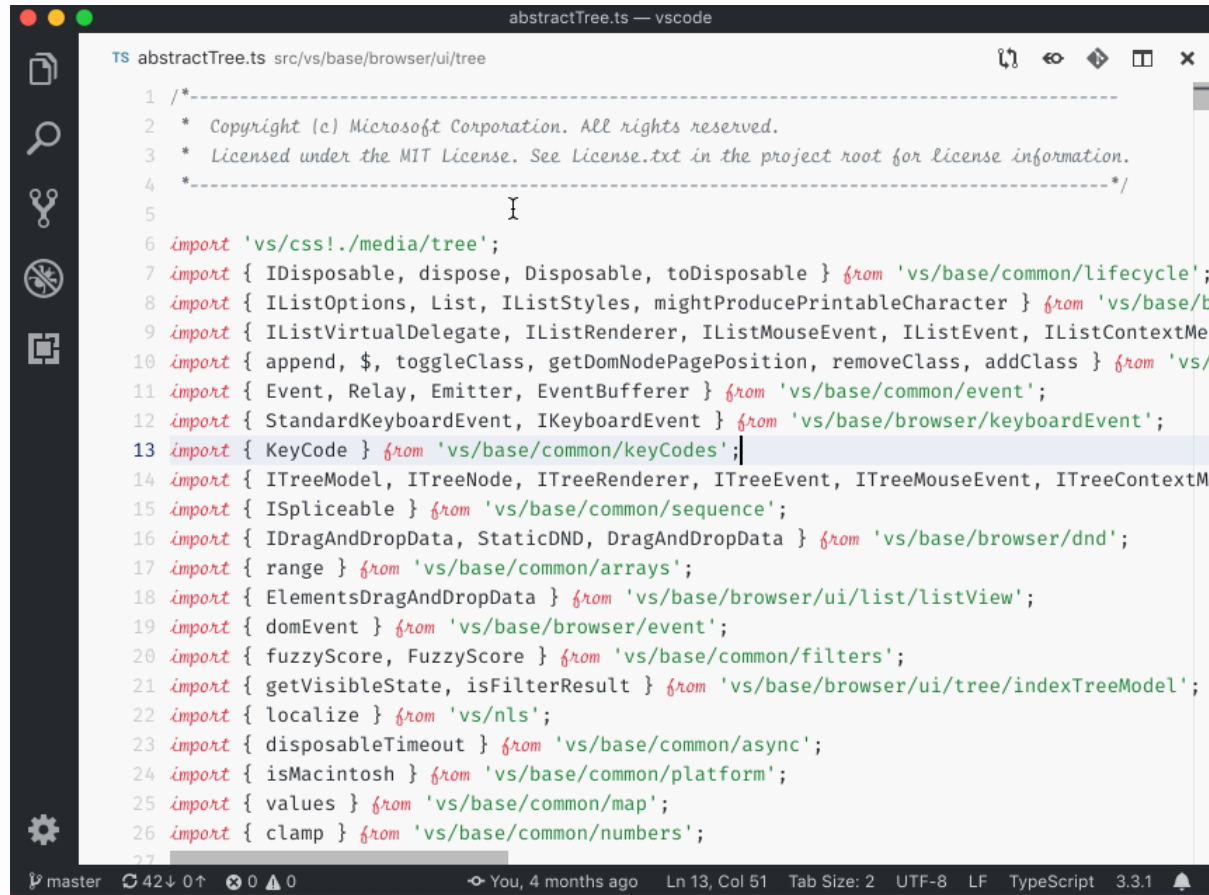
LogPoints

Non-breaking **log points** to show logging



Inbuilt Features

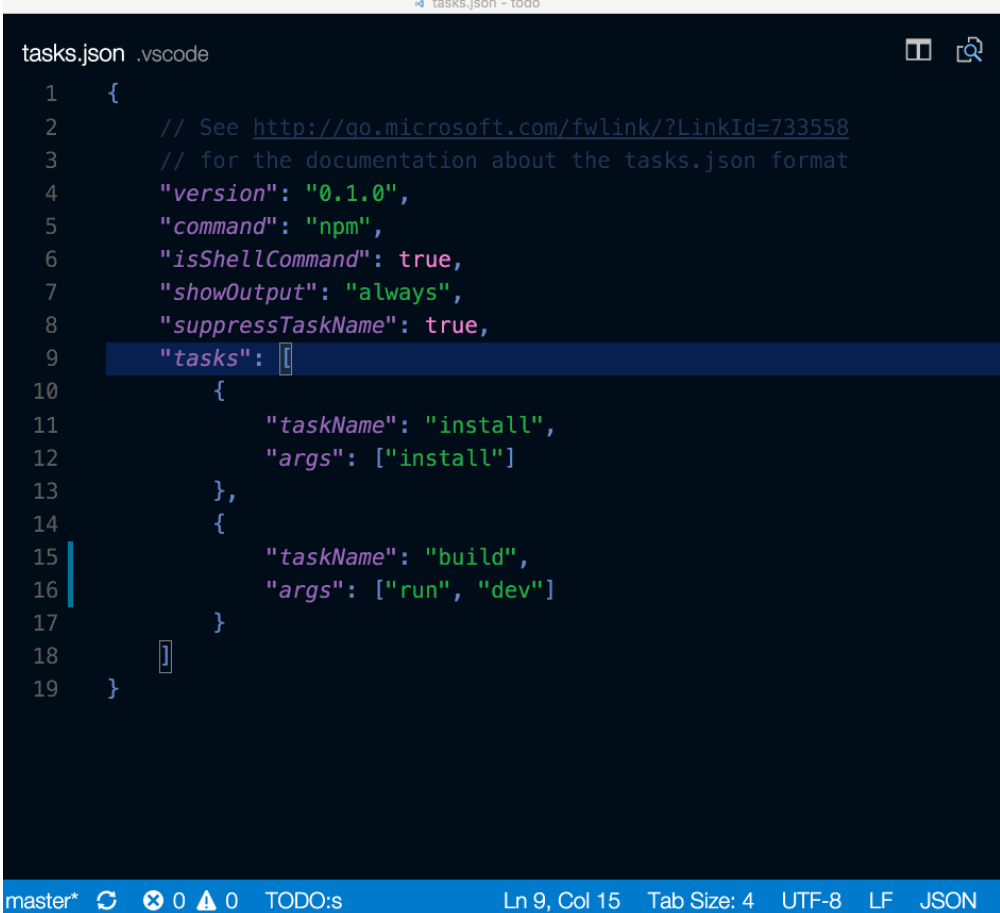
Screencast Mode



```
abstractTree.ts — vscode
TS abstractTree.ts src/vs/base/browser/ui/tree
1  /*-----
2   * Copyright (c) Microsoft Corporation. All rights reserved.
3   * Licensed under the MIT License. See License.txt in the project root for license information.
4   *-----*/
5
6  import 'vs/css!./media/tree';
7  import { IDisposable, dispose, Disposable, toDisposable } from 'vs/base/common/lifecycle';
8  import { IListOptions, List, IListStyles, mightProducePrintableCharacter } from 'vs/base/t
9  import { IListVirtualDelegate, IListRenderer, IListMouseEvent, IListEvent, IListContextMe
10 import { append, $, toggleClass, getDomNodePagePosition, removeClass, addClass } from 'vs/
11 import { Event, Relay, Emitter, EventBufferer } from 'vs/base/common/event';
12 import { StandardKeyboardEvent, IKeyboardEvent } from 'vs/base/browser/keyboardEvent';
13 import { KeyCode } from 'vs/base/common/keyCodes';
14 import { ITreeModel, ITreeNode, ITreeRenderer, ITreeEvent, ITreeMouseEvent, ITreeContextM
15 import { ISpliceable } from 'vs/base/common/sequence';
16 import { IDragAndDropData, StaticDND, DragAndDropData } from 'vs/base/browser/dnd';
17 import { range } from 'vs/base/common/arrays';
18 import { ElementsDragAndDropData } from 'vs/base/browser/ui/list/listView';
19 import { domEvent } from 'vs/base/browser/event';
20 import { fuzzyScore, FuzzyScore } from 'vs/base/common/filters';
21 import { getVisibleState, isFilterResult } from 'vs/base/browser/ui/tree/indexTreeModel';
22 import { localize } from 'vs/nls';
23 import { disposableTimeout } from 'vs/base/common/async';
24 import { isMacintosh } from 'vs/base/common/platform';
25 import { values } from 'vs/base/common/map';
26 import { clamp } from 'vs/base/common/numbers';
27
```

Task Runners

Run tasks like build, test and other custom tasks in **Task Runners**.



```
tasks.json .vscode
1  {
2    // See http://go.microsoft.com/fwlink/?LinkId=733558
3    // for the documentation about the tasks.json format
4    "version": "0.1.0",
5    "command": "npm",
6    "isShellCommand": true,
7    "showOutput": "always",
8    "suppressTaskName": true,
9    "tasks": [
10      {
11        "taskName": "install",
12        "args": ["install"]
13      },
14      {
15        "taskName": "build",
16        "args": ["run", "dev"]
17      }
18    ]
19  }
```

master* 0 0 TODO:s Ln 9, Col 15 Tab Size: 4 UTF-8 LF JSON

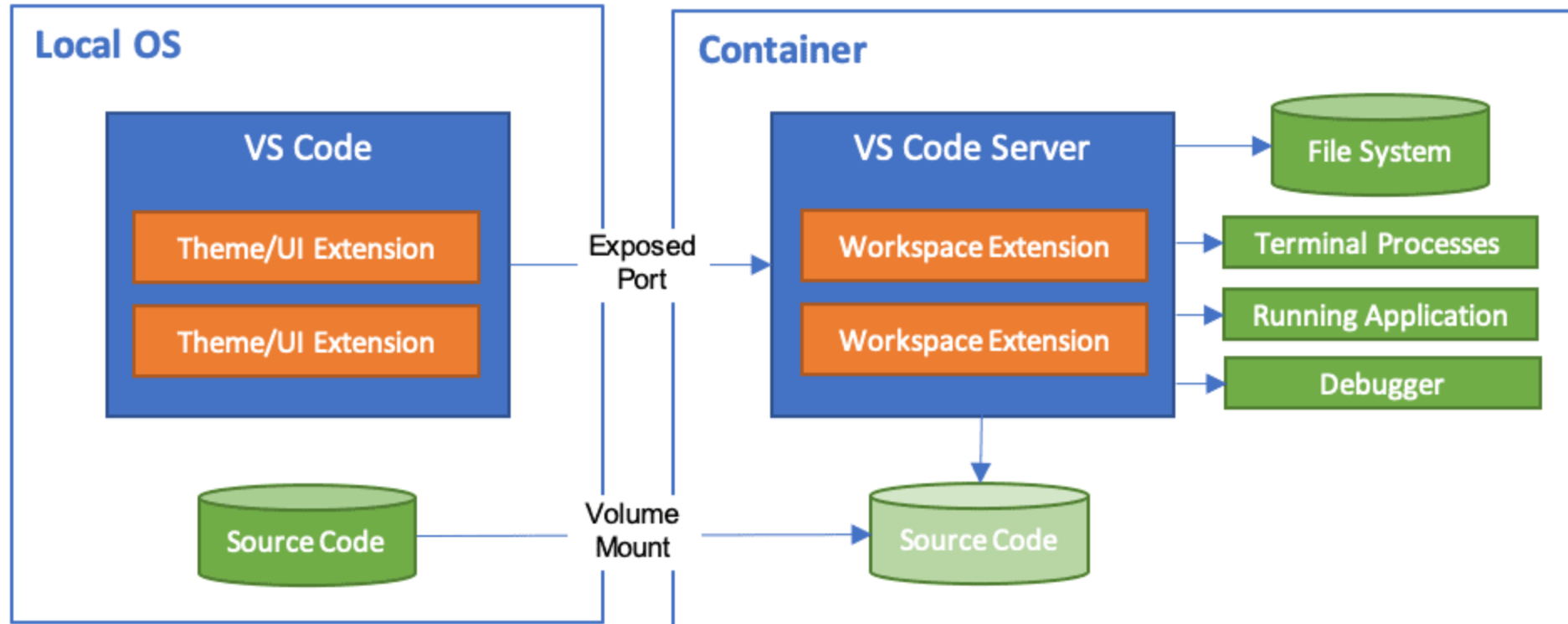
Remote Development

Remote	Description
Remote via SSH	Connect to remote machines with Visual Studio Code via SSH.
Work in WSL	Work in Windows Subsystem for Linux.
Develop in Containers	Work in a Docker Container.
GitHub Codespaces	Work in codespace with Visual Studio Code.

Dev Containers

Dev Containers uses a `devcontainer.json` file in your project tells VS Code how to access (or create) a development container with a well-defined tool and runtime stack.

You can use `prebuild images` to dev container setup. Also supports switching between `multiple containers`



Extensions

Installation

Install extensions with `Ctrl+Shift+X`. You can add to the workspace `@recommended` extensions for a project.

Most Popular extensions can be found with `@popular`

Extension Packs

Extension Packs

Extensions

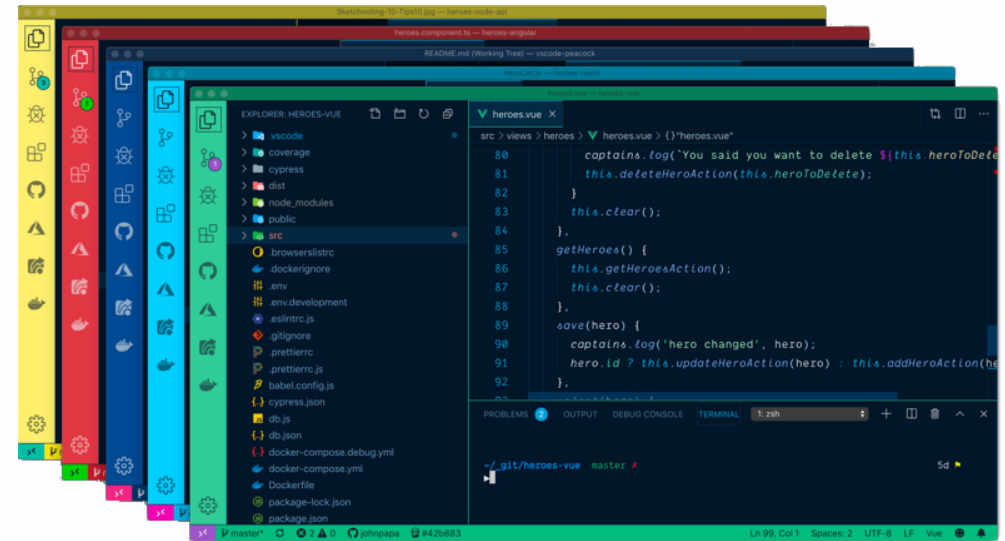
Marp

This slide deck was written in Markdown within VSCode. I used the [Marp for VS Code](#) extension to preview it.

[Marp](#) has a CLI - so this slide has a CI/CD pipeline 😊

Peacock

Peacock allows you to
colour code VS Code.

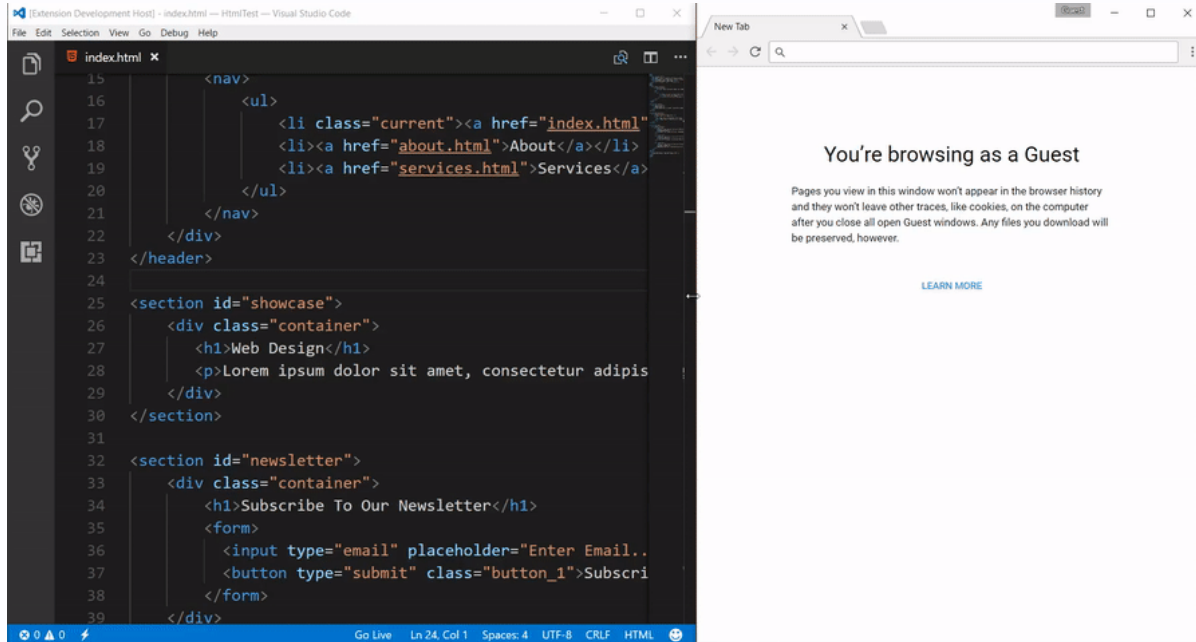


IntelliCode

AI-assisted
development features
for Python,
TypeScript/JavaScript
and Java developers

IntelliCode

```
loss = tf.reduce_sum(tf.square(linear_model - y))  
optimizer = tf.train.GradientDescentOptimizer(0.01)  
  
train = optimizer
```

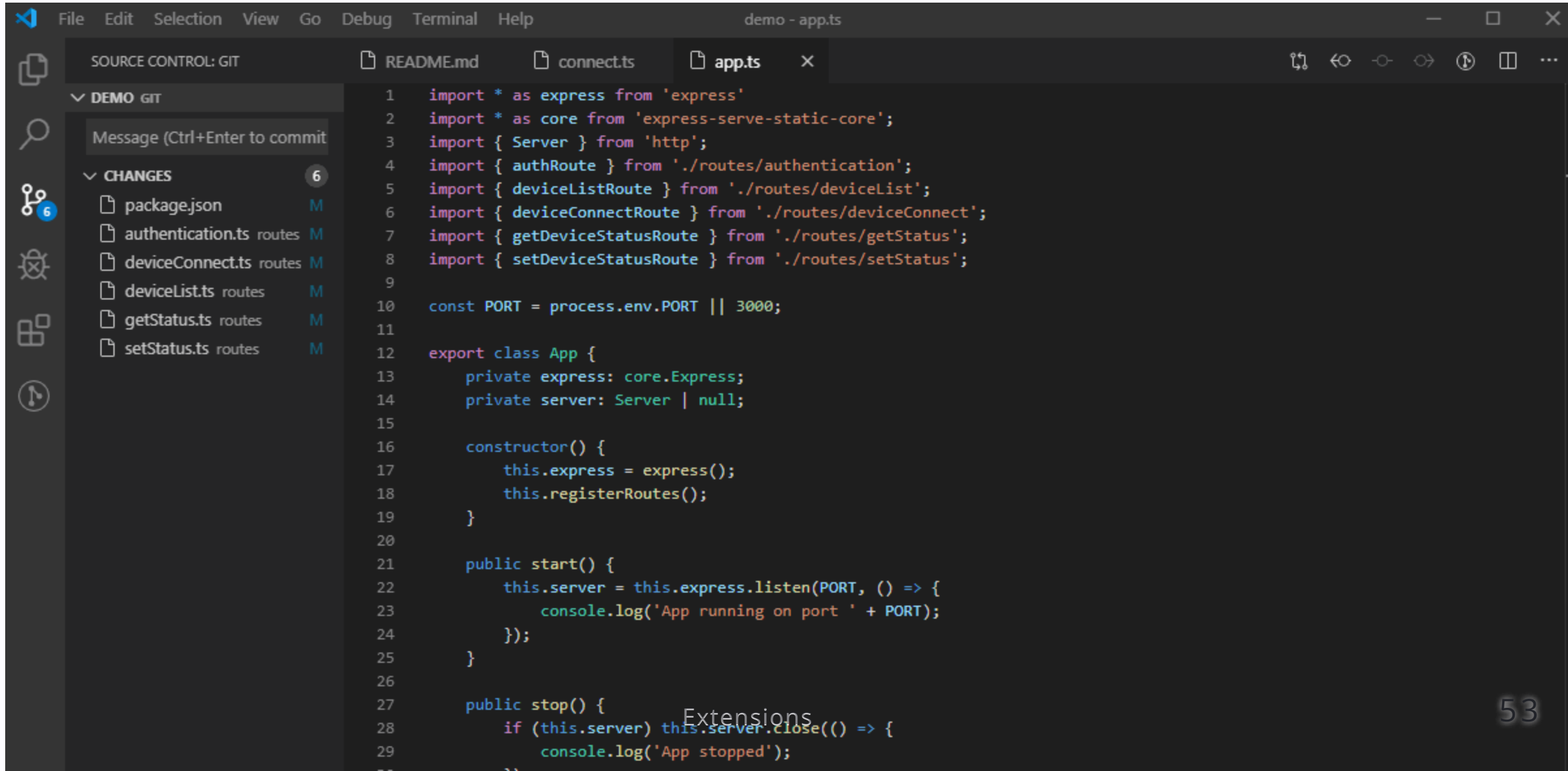


LiveServer

Launch a development
local Server with live
reload feature for
static & dynamic pages

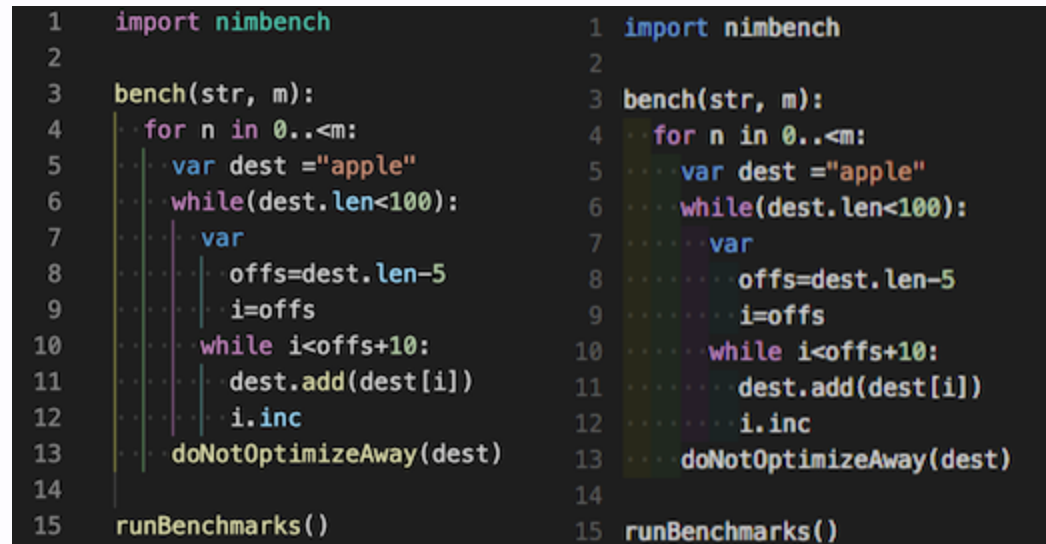
LiveServer

Git Graph creates a view of your repository, and easily perform Git actions from the graph.



Indent Rainbow

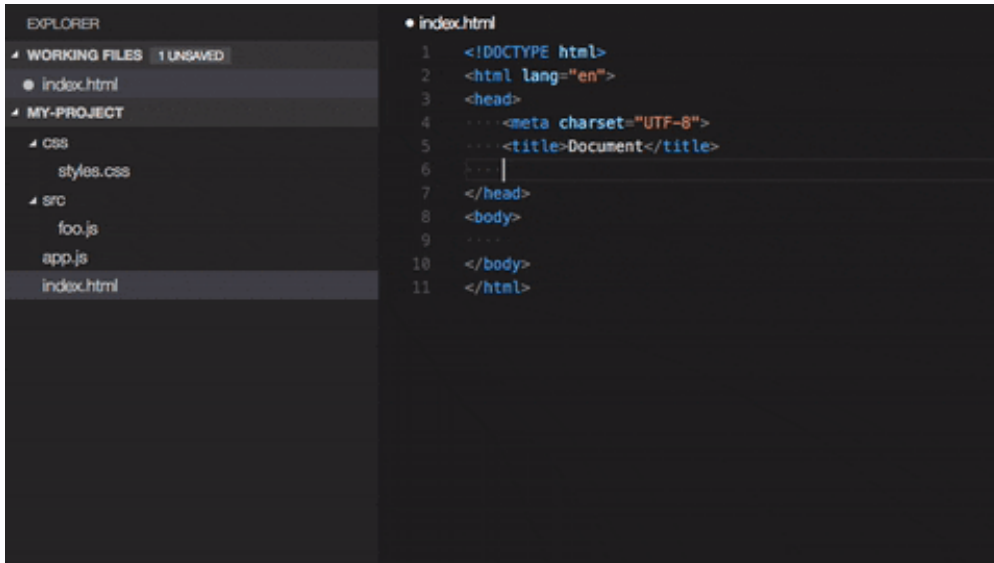
Indent Rainbow makes indentation more readable.
Highlights when you get indenting wrong.



The image shows two side-by-side code snippets. The left snippet is the original code with standard indentation, where nested loops and blocks are indicated by spaces. The right snippet is the same code with Indent Rainbow applied, which uses vertical colored lines to mark the start of each indentation level, making the structure much clearer. The colors used are green for the first level, blue for the second, and purple for the third.

```
1 import nimbench
2
3 bench(str, m):
4   for n in 0..<m:
5     var dest = "apple"
6     while(dest.len<100):
7       var
8         offs=dest.len-5
9         i=offs
10        while i<offs+10:
11          dest.add(dest[i])
12          i.inc
13        doNotOptimizeAway(dest)
14
15 runBenchmarks()
```

```
1 import nimbench
2
3 bench(str, m):
4   for n in 0..<m:
5     var dest = "apple"
6     while(dest.len<100):
7       var
8         offs=dest.len-5
9         i=offs
10        while i<offs+10:
11          dest.add(dest[i])
12          i.inc
13        doNotOptimizeAway(dest)
14
15 runBenchmarks()
```



Path Intellisense

Path Intellisense
autocompletes
filenames

Code Spell Checker

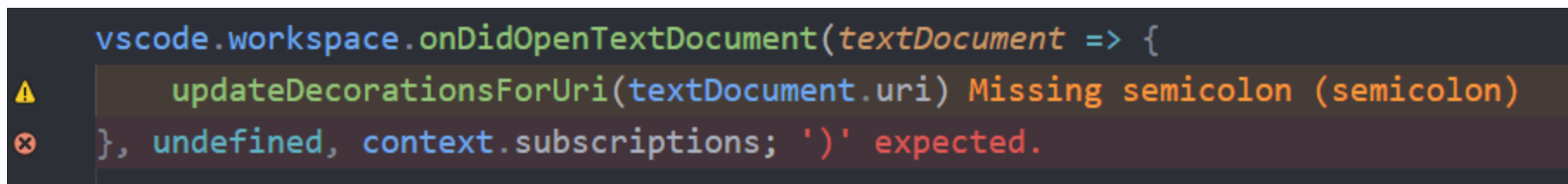
Code Spell Checker for
source code

```
• validator.ts src
1  import {
2    |   TextDocument, Diagnostic, DiagnosticSeverity,
3  } from 'vscode-languageserver';
4  import { isWordInDictionary } from './spellChecker';
5  import * as Text from './util/text';
6
7  import * as Rx from 'rx';
8  import { merge } from 'tsmerge';
9
10
11  /**
12   * default constants
13   */
14  const defaultMaxNumberOfProblems = 200;
15  const defaultMinWordLength      = 4;
16
17  export interface ValidationOptions {
18    |   maxNumberOfProblems?: number;
19    |   minWordLength?: number;
20    |   // words to always flag as an error
21    |   flagWords?: string[];
22  }
23
24  export function validateTextDocument(textDocument: TextDocument, options: ValidationOptions) {
25    |   return validateTextDocumentAsync(textDocument, options)
26    |   .toArray()
```

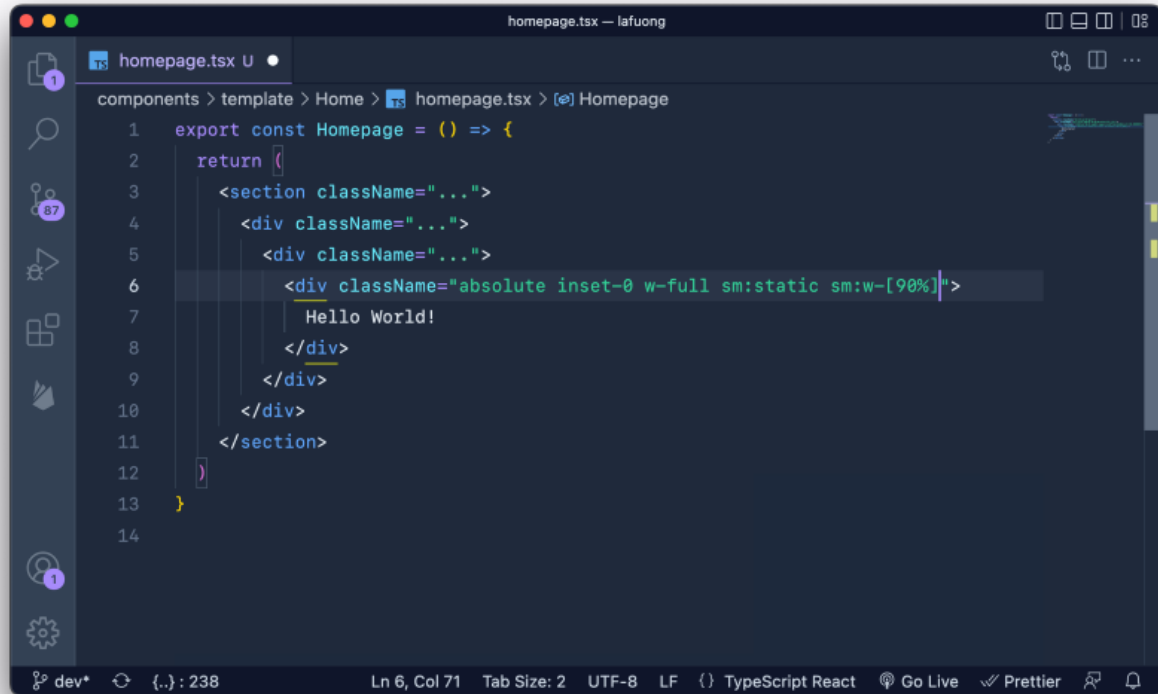
Error Lens

Improve highlighting of errors, warnings and other language diagnostics.

Error Lens

A screenshot of a code editor showing a TypeScript function definition. The function is named `vscode.workspace.onDidOpenTextDocument` and takes a parameter `textDocument => {`. The function body contains two lines: `updateDecorationsForUri(textDocument.uri)` and `}, undefined, context.subscriptions; ')' expected.`. The first line is highlighted with a yellow warning icon (a triangle with an exclamation mark) and the text "Missing semicolon (semicolon)". The second line is highlighted with a red error icon (a circle with an X) and the text "}, undefined, context.subscriptions; ')' expected.".

```
vscode.workspace.onDidOpenTextDocument(textDocument => {  
  updateDecorationsForUri(textDocument.uri) Missing semicolon (semicolon)  
}, undefined, context.subscriptions; ')' expected.
```



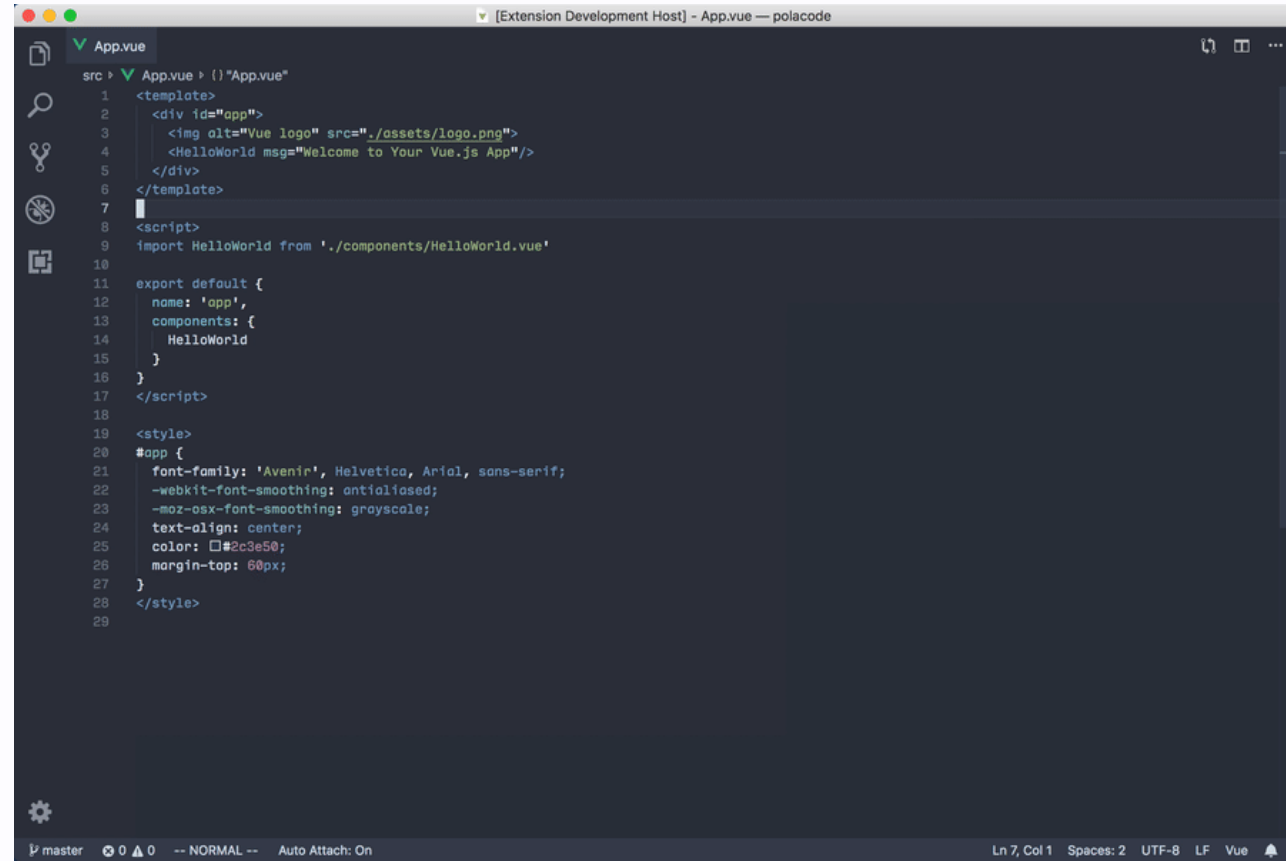
Inline Fold

Inline fold A custom decorator that "fold" matching content in single line. Helps with Tailwind CSS.

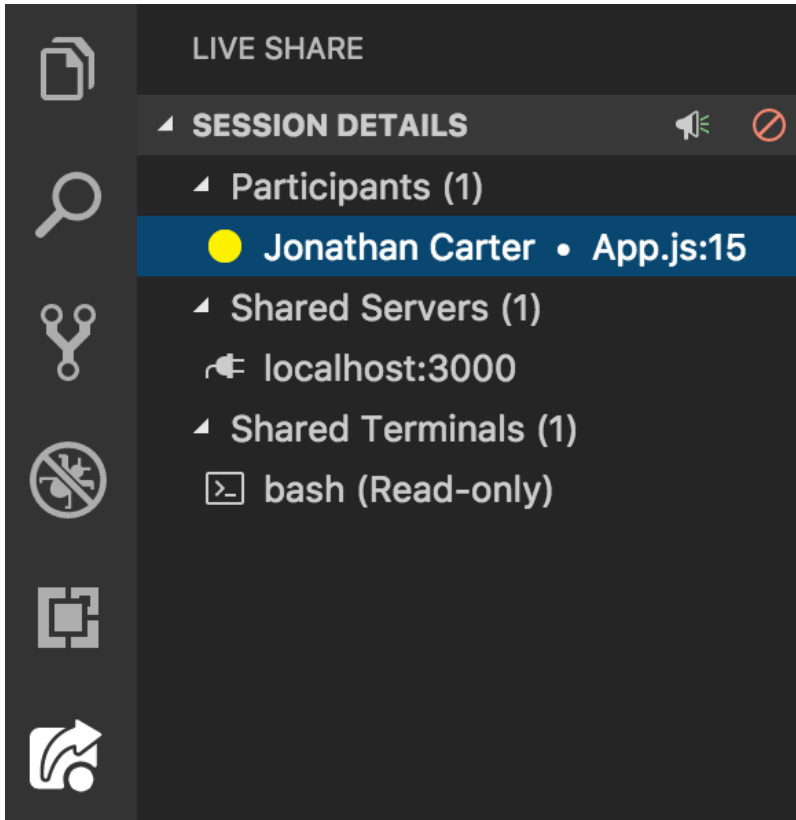
Polacode

 Polaroid for your code. Take screenshots of your code

Polacode



```
src > App.vue > () "App.vue"
1  <template>
2    <div id="app">
3      
4      <HelloWorld msg="Welcome to Your Vue.js App"/>
5    </div>
6  </template>
7
8  <script>
9    import HelloWorld from '../components/HelloWorld.vue'
10
11  export default {
12    name: 'app',
13    components: {
14      HelloWorld
15    }
16  }
17 </script>
18
19 <style>
20 #app {
21   font-family: 'Avenir', Helvetica, Arial, sans-serif;
22   -webkit-font-smoothing: antialiased;
23   -moz-osx-font-smoothing: grayscale;
24   text-align: center;
25   color: #2c3e50;
26   margin-top: 60px;
27 }
28 </style>
29
```



LiveShare

Real-time collaborative development from the comfort of your favourite tools.

[LiveShare](#) / [Docs](#)

Jira and Bitbucket

Bringing the power of Jira and Bitbucket to VS Code

- You can create and view issues
- Start work on issues
- Create pull requests
- Do code reviews
- Start builds
- Get build statuses and more!

Jira and Bitbucket

GitHub Copilot

GitHub Copilot Your AI
pair programmer

```
1
2 def common_prefix(a, b) :
3     """Return the common prefix of two lists."""
4     if len(a) < len(b) :
5         return common_prefix(b,a)
6     for i in range(len(a)) :
7         if a[i] != b[i] :
8             return a[:i]
9     return a
10
11
12
```

RooCode

RooCode is whole dev team of AI agents in your editor. Supports multiple LLM providers and you can choose the models easily.

C# Dev Kit

Official C# extension from Microsoft

- Project System / Solution Explorer
- Code Editing (Uses the C# extension)
- Package Management
- Debugging
- Testing

C# Dev Kit

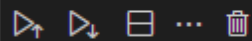
Markdown Example

```
Console.WriteLine("C# Example");
```

✓ 0.0s

csharp - C# Script Code

C# Example



```
gantt
```

```
... title A Gantt Diagram
... dateFormat YYYY-MM-DD
... section Section
... A task .....:a1, 2024-01-01, 1d
... Another task .....:after a1, 2d
... section Another
... Task in sec .....:2024-01-04, 3d
... another task .....: 2d
```

✓ 0.0s

mermaid - Mermaid Code



```
<h1>HTML Example</h1>
```

✓ 0.0s

html - HTML Code

HTML Example

Polygot Notebooks

Polygot Notebooks
allows

C#, F#, PowerShell,
JavaScript, SQL, KQL,
Python, R, HTML,
HTTP, Mermaid.

API Clients

REST Client

```
GET https://example.com/topics/1 HTTP/1.1
```

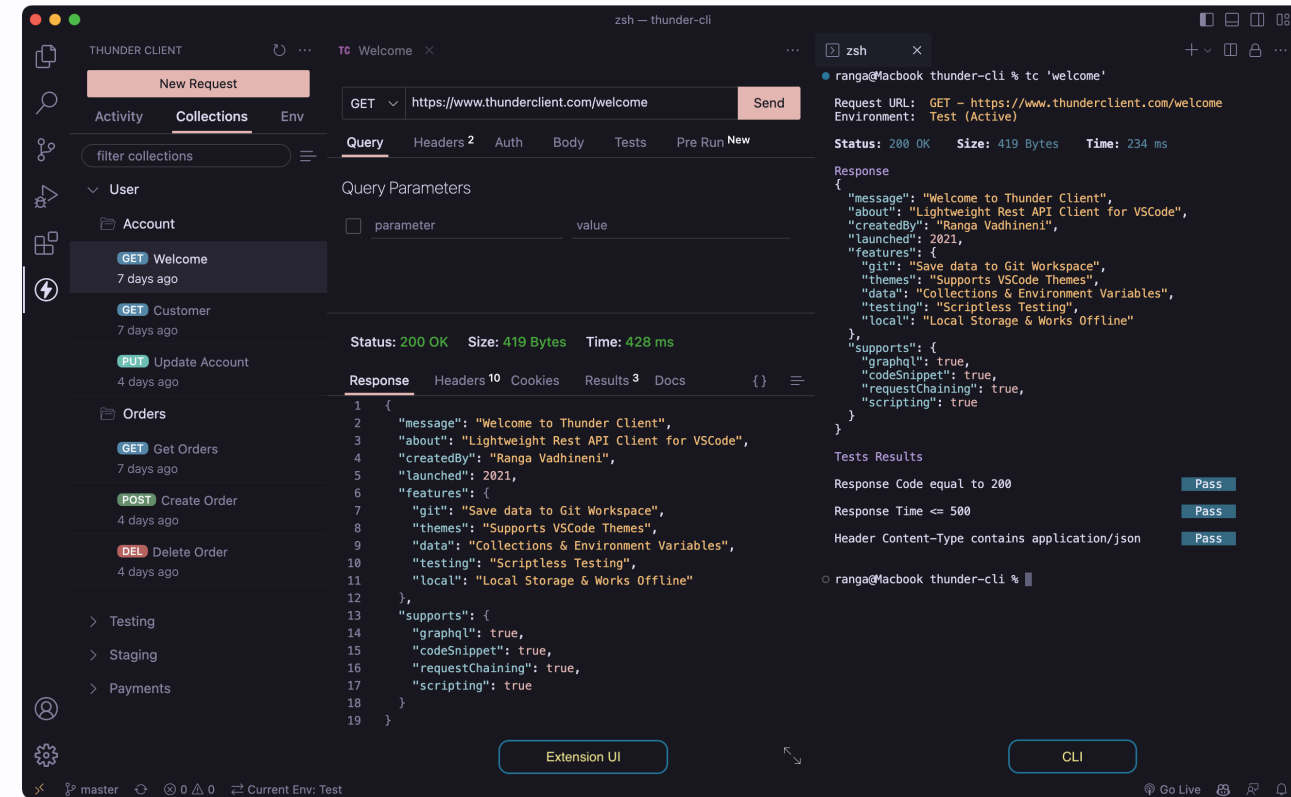
```
POST https://example.com/comments HTTP/1.1  
content-type: application/json
```

```
{  
  "name": "sample",  
  "time": "Wed, 21 Oct 2015 18:27:50 GMT"  
}
```

```
curl https://www.example.com
```

Thunder Client

Lightweight Rest API Client for VS Code



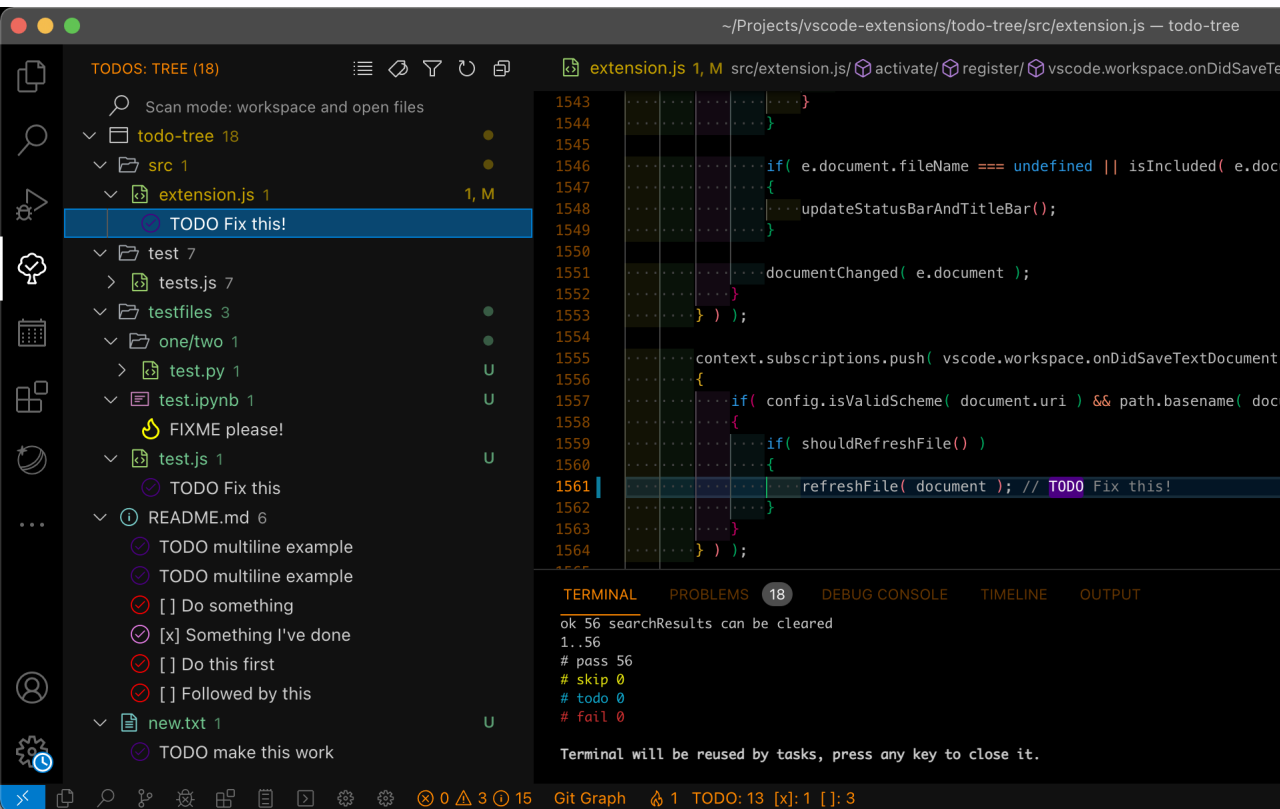
```

2  export class MyClass {
3
4      /**
5       * MyMethod
6       * * Important information is highlighted
7       * ! Deprecated method, do not use
8       * ? Should this method be exposed in the public API?
9       * TODO: refactor this method so that it conforms to the API
10      * @param myParam The parameter for this method
11      */
12     public MyMethod(myParam: any): void {
13         let myVar: number = 123;
14
15         /** This is highlighted
16         if (myVar > 0) {
17             throw new TypeError(); ///! This is an alert
18         }
19
20         ///? This is a query
21         let x = 1;
22
23         //// this.lineOfCode() == commentedOut;
24
25         //TODO: Create some test cases
26     }
27 }

```

Better Comments

Better Comments
colour codes your
commit messages

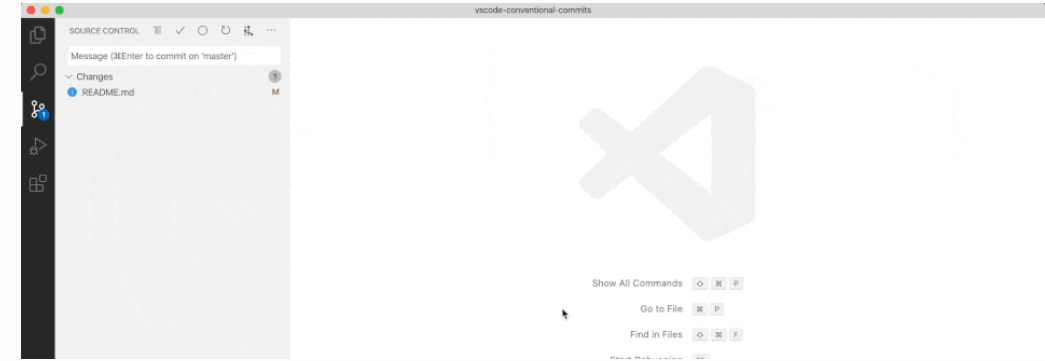


TODO Tree
TODO Tree quickly find
all your todo items.

Conventional Commits

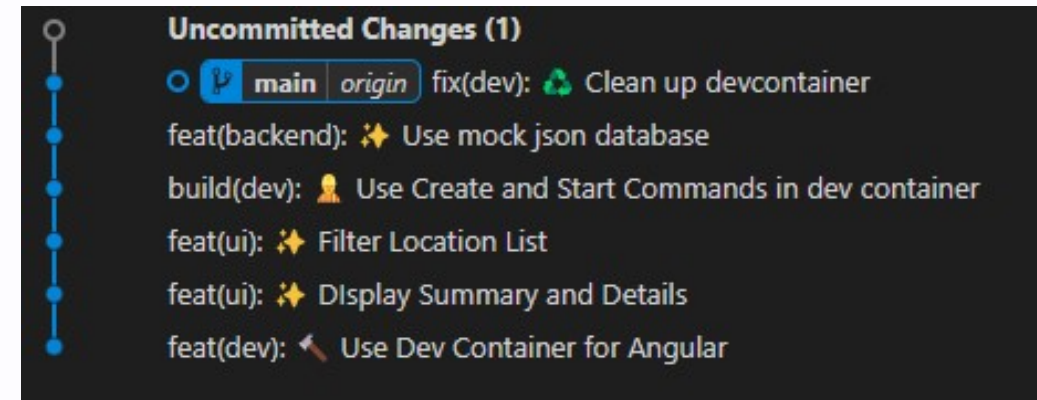
Conventional Commits

```
<type>[optional scope]: <description>  
[optional body]  
[optional footer(s)]
```



Follows
conventionalcommits.org
and uses [gitmoji](https://gitmoji.dev) which then
dovetails to semver.org

```
feat(api): ✨ Add Get User Endpoint  
Allow users to access their profile  
ABC-123
```



Unique Lines

Keep **unique lines** of text and remove duplicates from current selection. Also includes a command to shuffle currently selected lines.

```
A      A
B =>   B
B      C
C
C
```

Encode Decode

Encode Decode converts text.

- Convert String **to/from** Base64, HTML Entities, JSON Byte Array, JSON String, Unicode, XML Entities
- Convert String to MD5/SHA1/SHA256/SHA512 (as Base64 or Hex)

```
extension.ts src/extension.ts activate/ vscode.workspace.onDidChange
48     return vscode.window.createTextEditorDecorationType(decorationOp
49 }
50
51 // this method is called when vs code is activated
52 export function activate(context: vscode.ExtensionContext) {
53
54     let bookmarks: BookmarksController;
55     let activeEditorCountLine: number;
56     let timeout: NodeJS.Timer;
57
58     const provider      BookmarksContentProvider();
59     const viewer = new WhatsNewManager(context).registerContentProvi
60     viewer.showPageInActivation();
61     context.subscriptions.push(vscode.commands.registerCommand("book
62
63     // load pre-saved bookmarks
64     const didLoadBookmarks: boolean = loadWorkspaceState();
65
```

Bookmarks

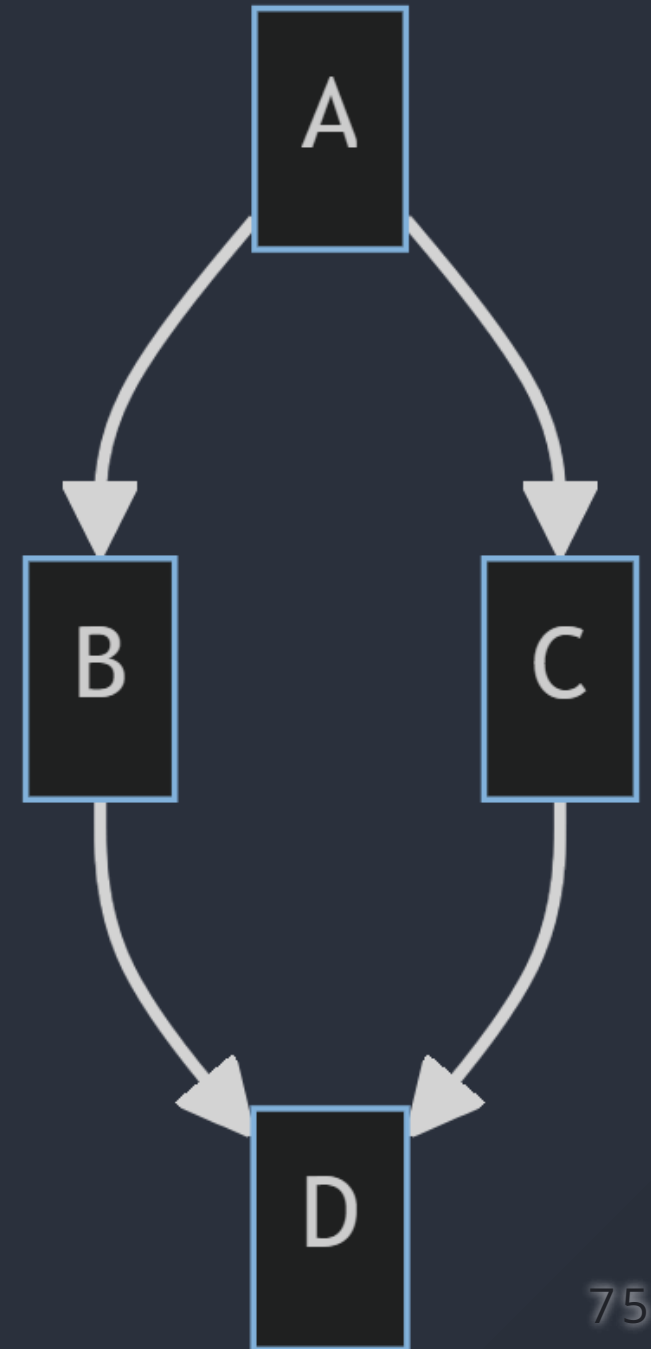
Bookmark your code
and quickly jump to
bookmarks.

Markdown Preview Mermaid Support

Markdown Preview Mermaid
Support allows you to create
Mermaid JS documents that can be
included in Markdown documents.

```
graph TD;
  A-->B;
  A-->C;
  B-->D;
  C-->D;
```

Extensions



Aside: Mermaid JS can generate

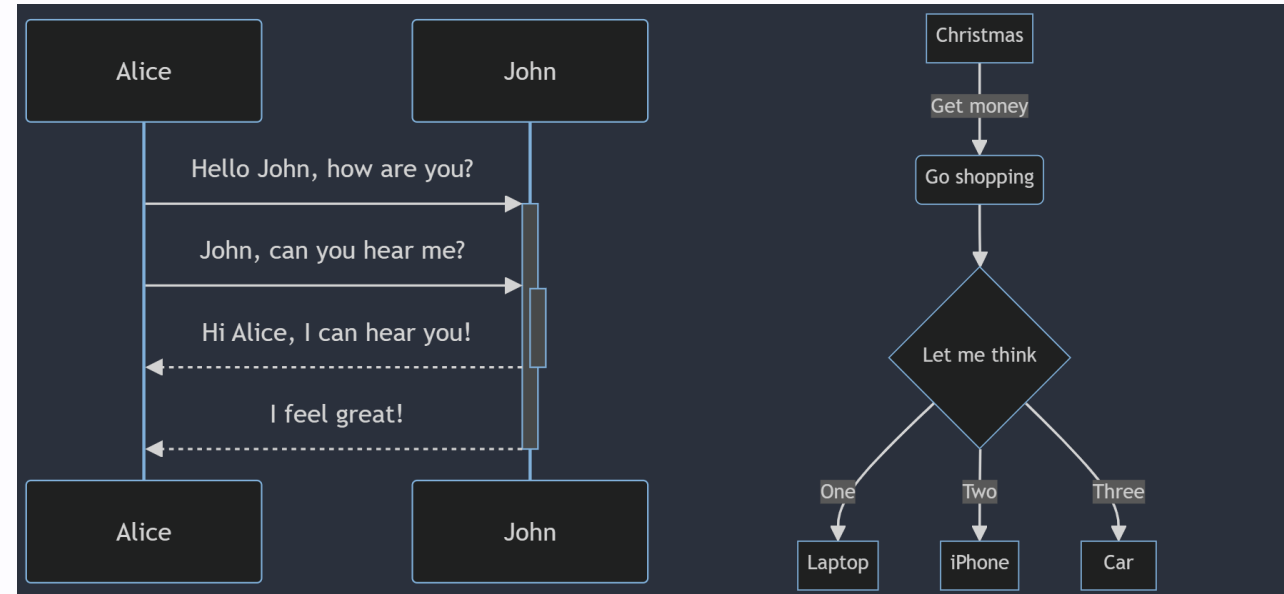
- Generate Flowchart
- Sequence Diagram
- Class Diagram
- State Diagram
- Entity Relationship Diagram
- User Journey
- Gantt
- Pie Chart
- Quadrant Chart
- Requirement Diagram

sequenceDiagram

```
Alice->>+John: Hello John, how are you?  
Alice->>+John: John, can you hear me?  
John-->>-Alice: Hi Alice, I can hear you!  
John-->>-Alice: I feel great!
```

flowchart TD

```
A[Christmas] -->|Get money| B(Go shopping)  
B --> C{Let me think}  
C -->|One| D[Laptop]  
C -->|Two| E[iPhone]  
C -->|Three| F[fa:fa-car Car]
```



```
heroes.json U x
heroes.json > [ ] members > { } 0 > [ ] powers
1 {
2   "squadName": "Super hero squad",
3   "homeTown": "Metro City",
4   "formed": 2016,
5   "secretBase": "Super tower",
6   "active": true,
7   "members": [
8     {
9       "name": "Molecule Man",
10      "age": 29,
11      "secretIdentity": "Dan Jukes",
12      "powers": [
13        "Radiation resistance",
14        "Turning tiny",
15        "Radiation blast"
16      ]
17    },
18    {
19      "name": "Aykut Saraç",
20      "age": 21,
21      "secretIdentity": "John Doe",
22      "powers": [
23        "React",
24        "TypeScript",
25        "JavaScript"
26      ]
27    }
28  ]
29 }
```

JSON Crack

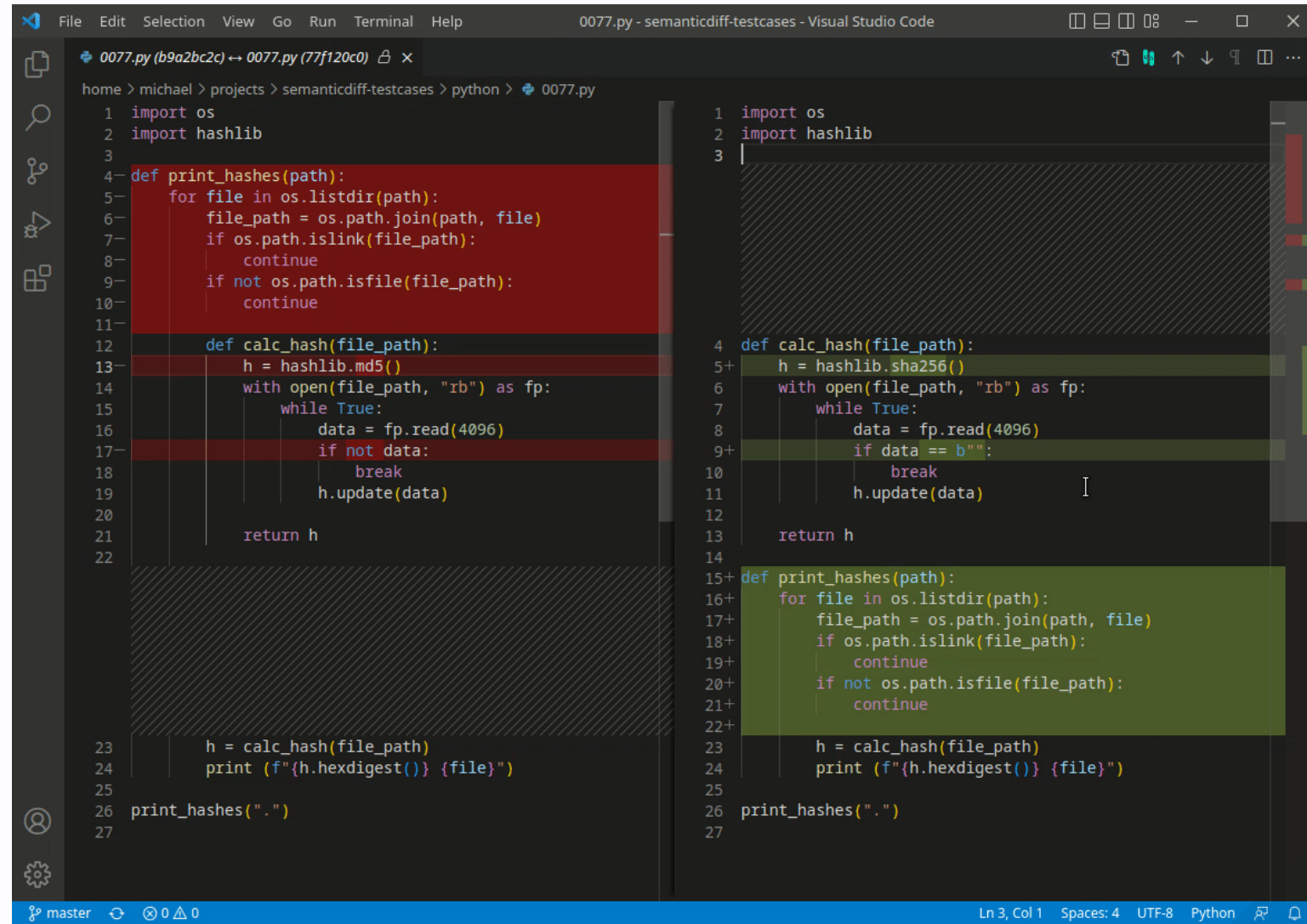
JSON Crack
seamlessly
visualize your
JSON data
instantly into
graphs.

SemanticDiff

SemanticDiff is a programming language aware diffs.

Can show differences when you move code.

Extensions



```
0077.py (b9a2bc2c) ↔ 0077.py (77f120c0)
home > michael > projects > semanticdiff-testcases > python > 0077.py

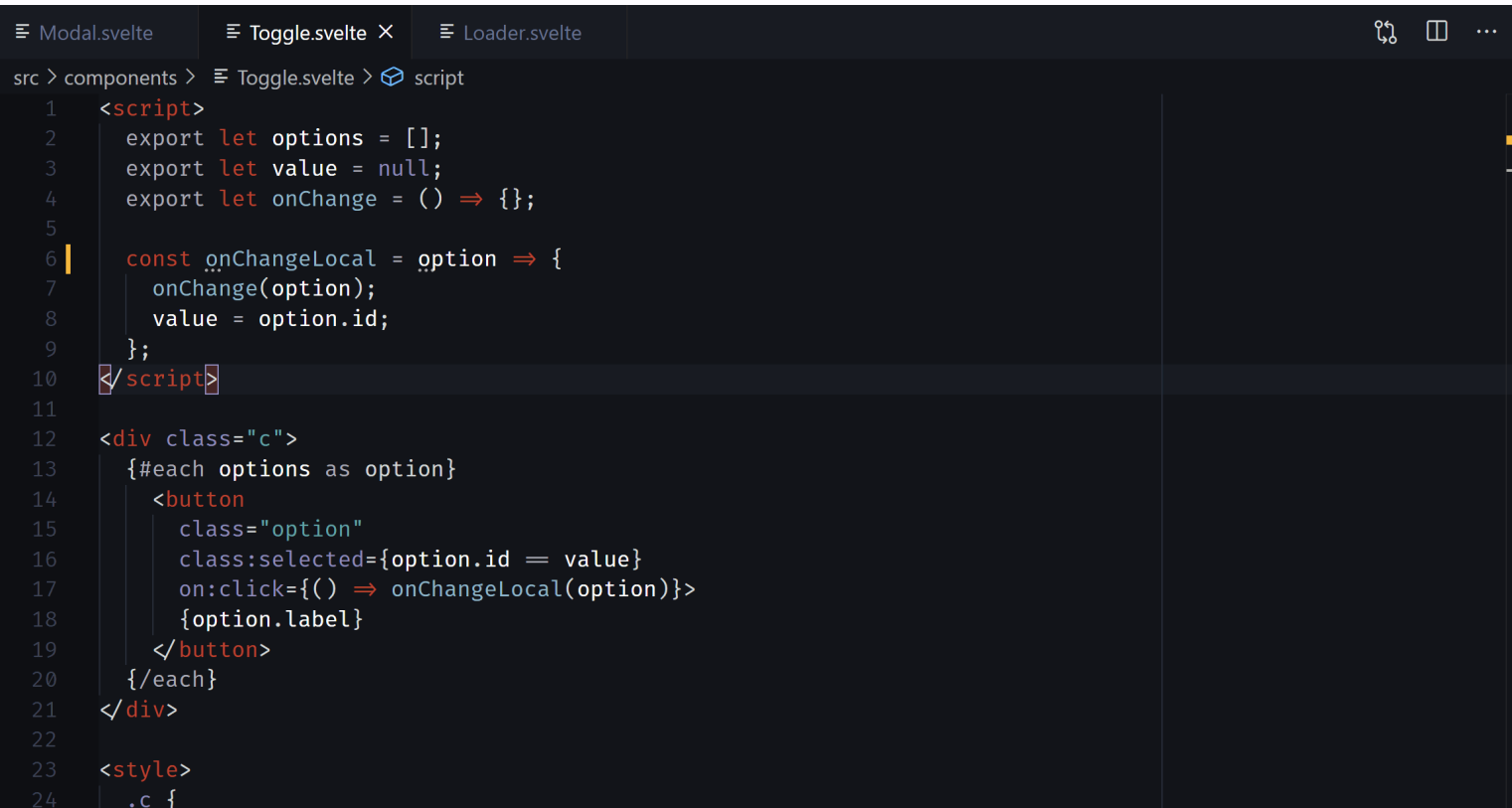
1 import os
2 import hashlib
3
4 def print_hashes(path):
5     for file in os.listdir(path):
6         file_path = os.path.join(path, file)
7         if os.path.islink(file_path):
8             continue
9         if not os.path.isfile(file_path):
10            continue
11
12     def calc_hash(file_path):
13         h = hashlib.md5()
14         with open(file_path, "rb") as fp:
15             while True:
16                 data = fp.read(4096)
17                 if not data:
18                     break
19                 h.update(data)
20
21         return h
22
23 h = calc_hash(file_path)
24 print (f"{h.hexdigest()} {file}")
25
26 print_hashes(".")
27
```

```
1 import os
2 import hashlib
3
4 def calc_hash(file_path):
5     h = hashlib.sha256()
6     with open(file_path, "rb") as fp:
7         while True:
8             data = fp.read(4096)
9             if data == b"":
10                 break
11             h.update(data)
12
13     return h
14
15 def print_hashes(path):
16     for file in os.listdir(path):
17         file_path = os.path.join(path, file)
18         if os.path.islink(file_path):
19             continue
20         if not os.path.isfile(file_path):
21             continue
22
23     h = calc_hash(file_path)
24     print (f"{h.hexdigest()} {file}")
25
26 print_hashes(".")
27
```

FootSteps

FootSteps

Highlight and
navigate
between your
most recently
edited chunks of
code

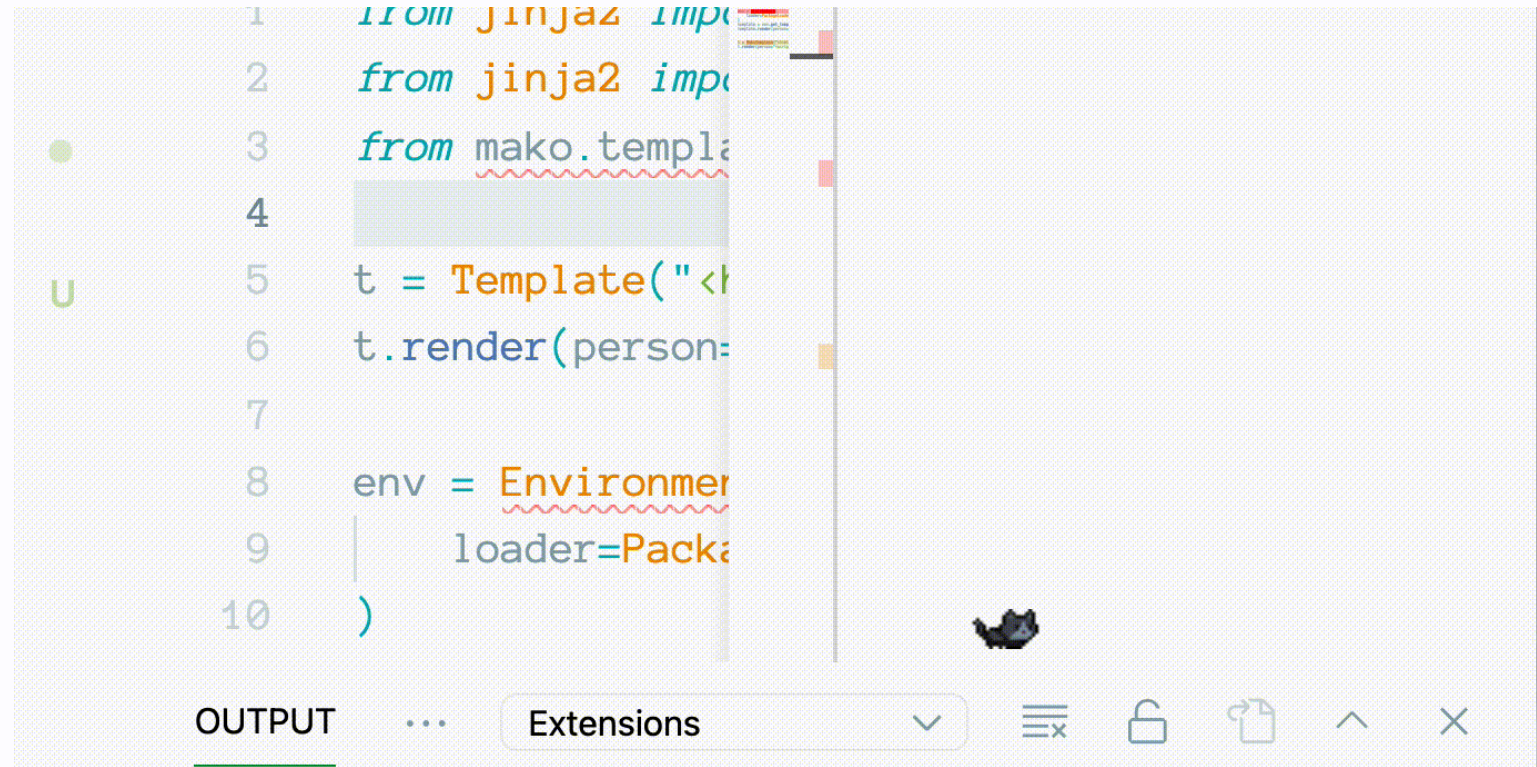


The screenshot shows a code editor with three tabs: Modal.svelte, Toggle.svelte (active), and Loader.svelte. The active tab displays Svelte code. A `<script>` block on lines 1-10 is highlighted in blue. The code includes exports for `options`, `value`, and `onChange`, and a local `onChange` function. Below the script is a `<div>` with a `{#each options as option}` loop containing a `<button>` that calls `onChangeLocal(option)` on click. The editor interface includes a breadcrumb 'src > components > Toggle.svelte > script' and line numbers 1 through 24.

```
1 <script>
2   export let options = [];
3   export let value = null;
4   export let onChange = () => {};
5
6   const onChangeLocal = option => {
7     onChange(option);
8     value = option.id;
9   };
10 </script>
11
12 <div class="c">
13   {#each options as option}
14     <button
15       class="option"
16       class:selected={option.id = value}
17       on:click={() => onChangeLocal(option)}>
18       {option.label}
19     </button>
20   {/each}
21 </div>
22
23 <style>
24   .c {
```

Pets for your VS Code

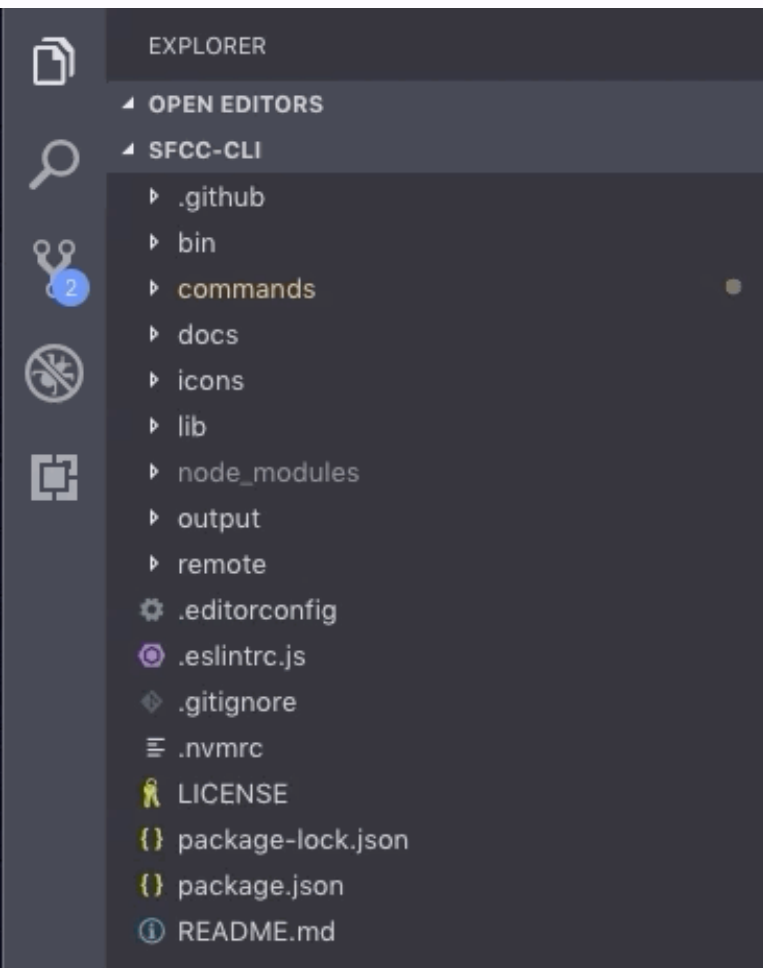
Pets gives you a
virtual pet to
play with while
you are coding.



The screenshot shows a VS Code editor window with a Python file. The code is as follows:

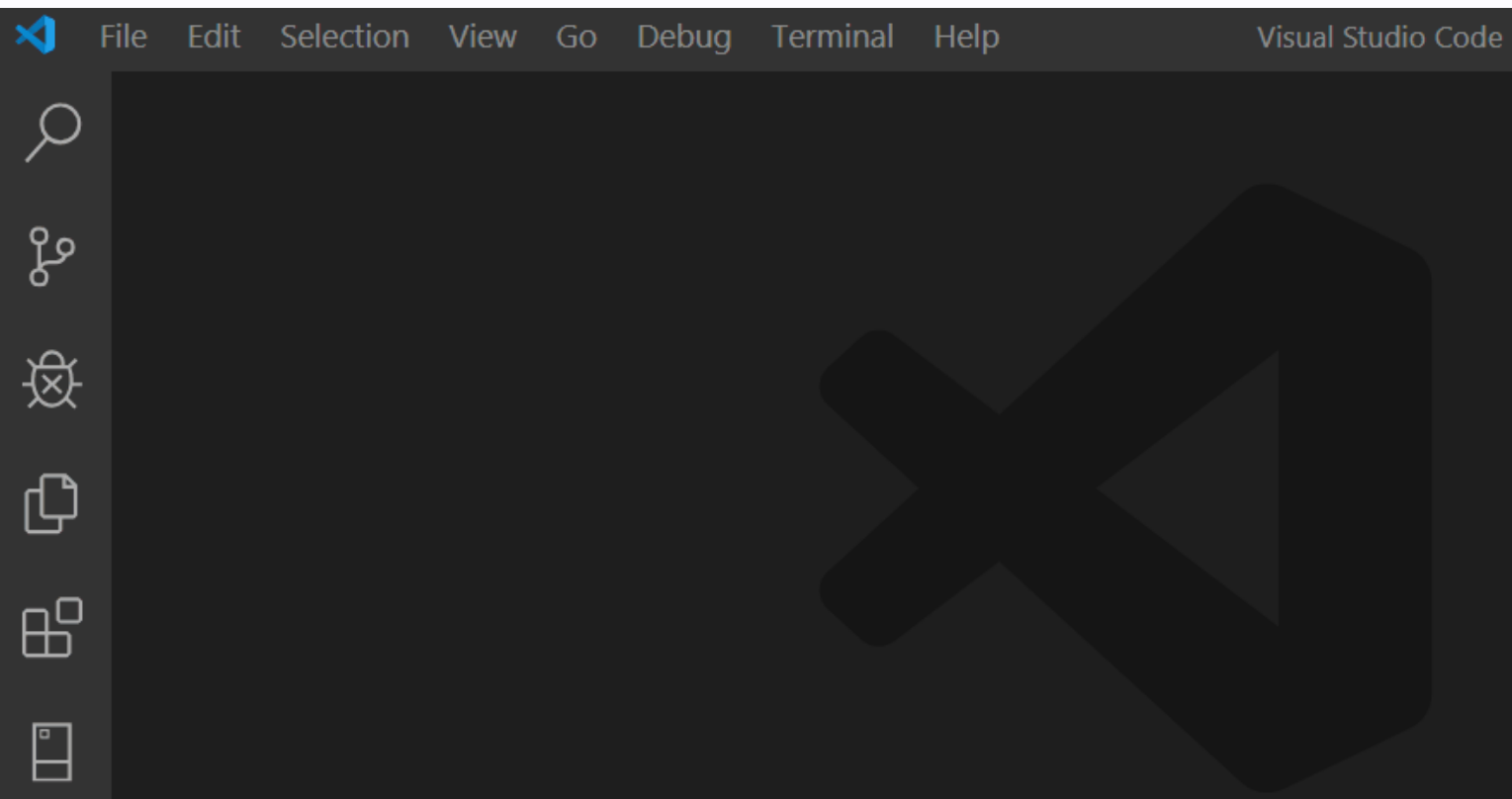
```
1 from jinja2 import
2 from jinja2 import
3 from mako.template
4
5 t = Template("<h
6 t.render(person=
7
8 env = Environment
9 | loader=Packa
10 )
```

The bottom status bar shows the 'OUTPUT' tab selected, followed by a menu icon, the 'Extensions' view, and several other icons (list, lock, copy, zoom in, zoom out).



Explorer Exclude

Hide and show files with **Explorer Exclude**. Persists to your `.vscode/settings.json` file under `files.exclude`.



Database Connections

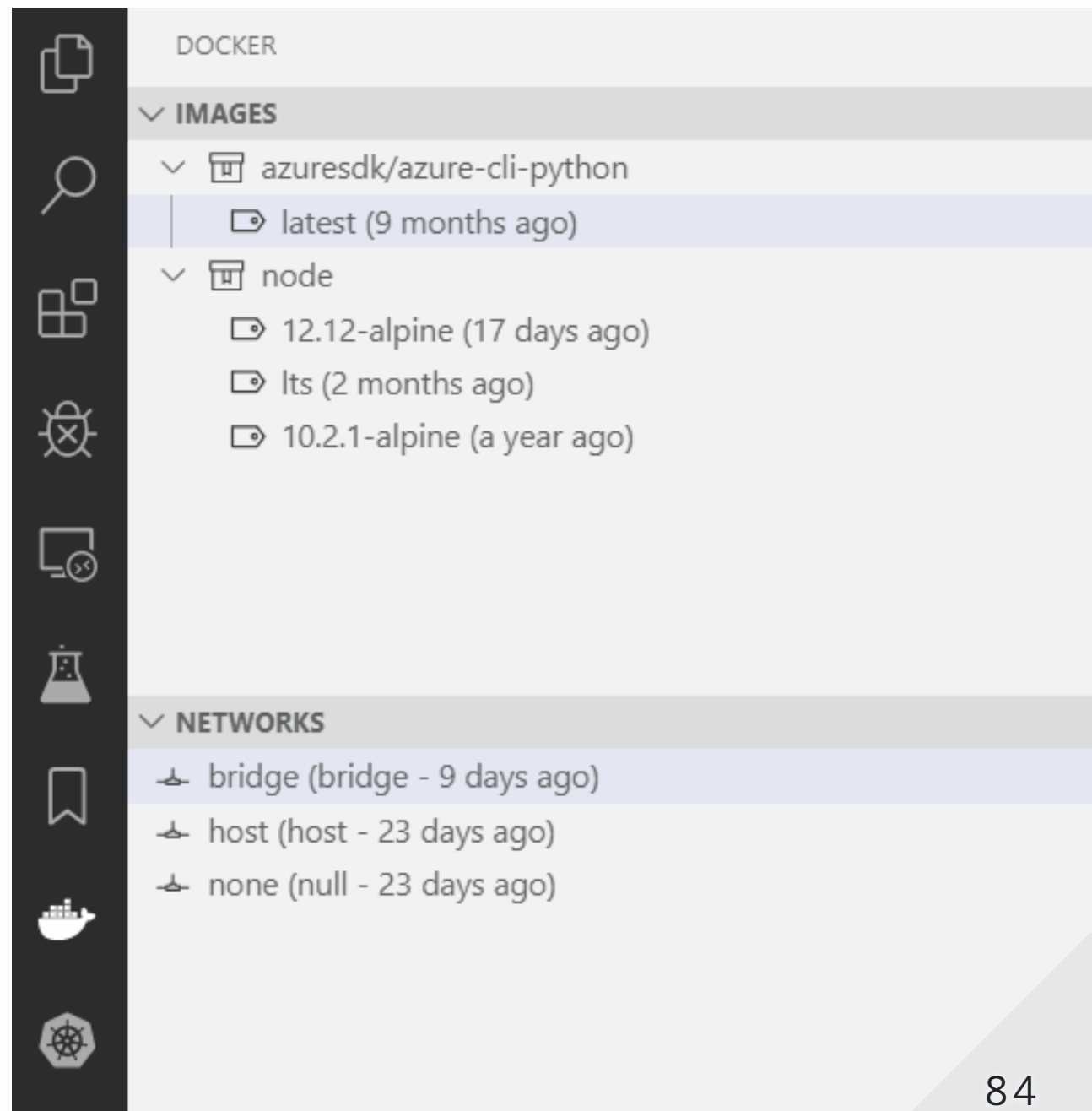
- SQL Server
- MySQL
- PostgreSQL
- MongoDB for VS Code
- Redis

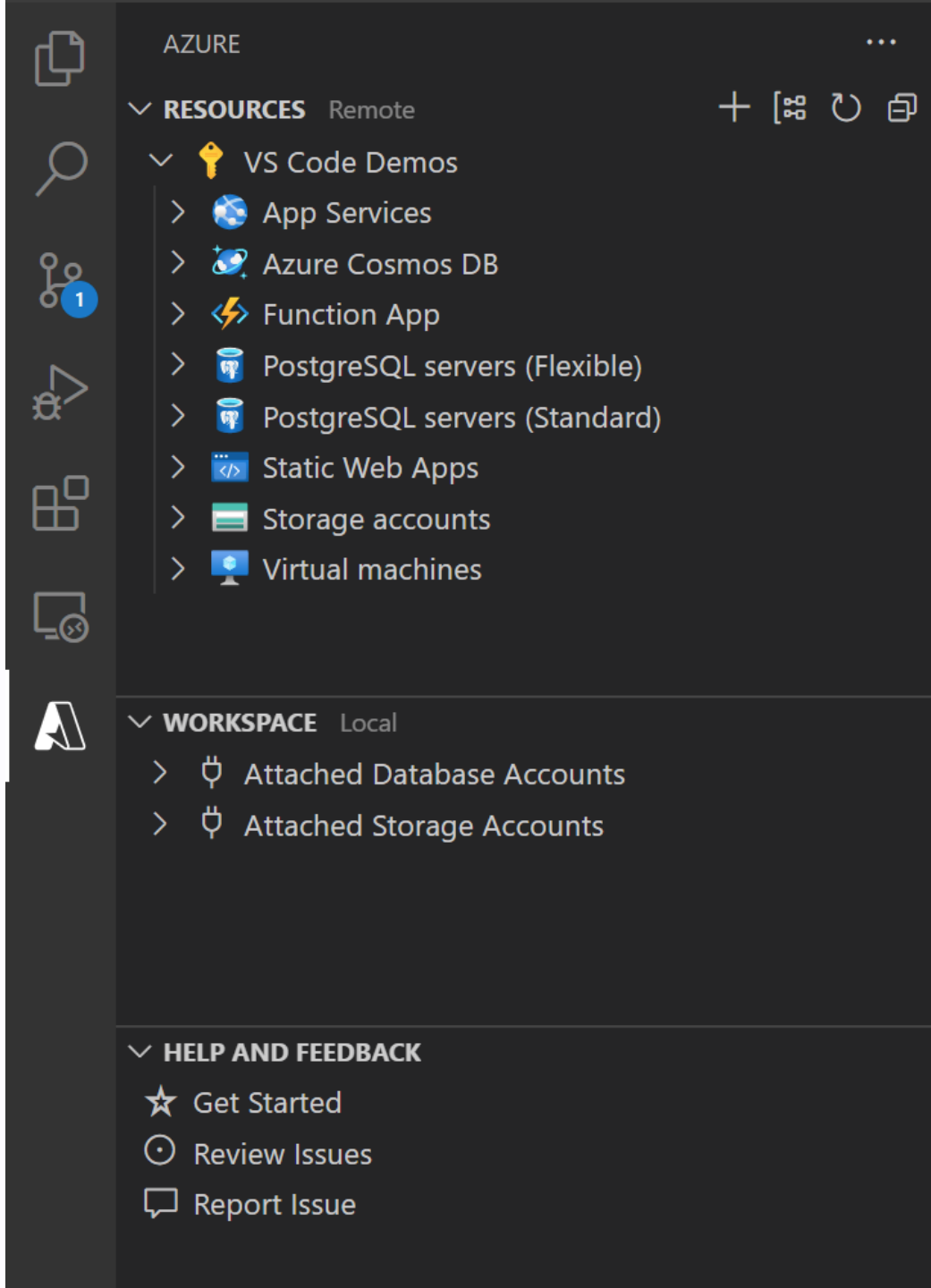
Docker

Makes it easy to
create, manage, and
debug containerized
applications.

Docker

Extensions





Azure Tools

Get web site hosting,
SQL and MongoDB
data, Docker
Containers, Serverless
Functions and more,
all on Azure.

Azure Tools

Extensions

AWS Toolkit

Including
CodeWhisperer,
CodeCatalyst, and
support for Lambda,
S3, CloudWatch Logs,
and many other
services

AWS Toolkit

Extensions

